17-9-2020 Thursday

Module-II

Analysis of First and Second Order Dynamic Circuits

Whenever, a circuit is switched from one condition to another either by a change in the applied source or a change in the circuit element, there is a transition period, during which the branch current & element voltages change from their former value to the new value. This period is called transient. After the transient has passed, the circuit is said to be in steady state.

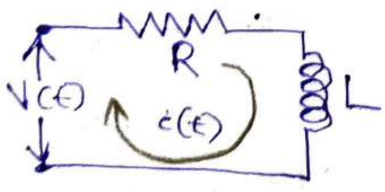
$$\boxed{\begin{aligned} V_L(t) &= L\frac{di}{dt} \\ i_c(t) &= C\frac{dV_c}{dt} \end{aligned}}$$

First order differential equation.

Complementary Function → transient

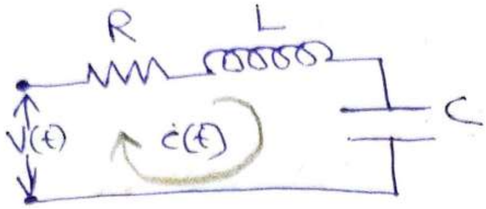
Particular integral → steady state

Consider a combination of R & L circuit element



$$\boxed{V(t) = L\frac{di(t)}{dt} + Ri(t)}$$

Drawing a RLC circuit



$i = \frac{dq}{dt}$

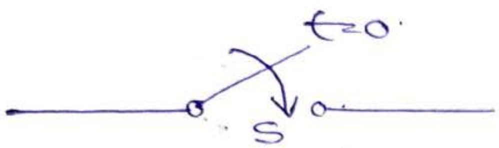
$$\boxed{\begin{aligned} V(t) &= R\,i(t) + L\frac{di}{dt} + \frac{1}{C}\int i\cdot dt \\ V(t) &= R\frac{dq}{dt} + L\frac{d^2q}{dt^2} + \frac{1}{C}q \end{aligned}}$$

2nd Order Differential Equation

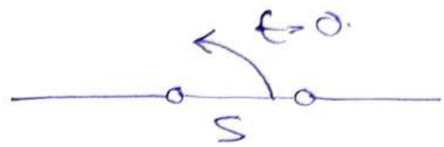
If the circuit contains a combination of Resistor & Inductor or combination of resistor & Capacitor, that circuit will be of the 1st order combination. Similarly, if we take a circuit with a combination of Inductor & Capacitor, then that circuit equation will be 2nd order.

The Voltage current for an Inductor or Capacitor is differential. The circuit containing an inductor 'L' or a capacitor 'C' and resistor will have current and Voltage Variables given by Differential Equations. If it is a linear first Order differential Equation with Constant co-efficients when the Value RLC are constant. Circuit having 2 storage elements like L & C are reffered to 2nd Order Circuits. the circuits change an ...

at time $(t) = 0$ & is represented by a switch.



Switch is closed at $t=0$.



Switch is opened at $t=0$.

IMP PT : Switching "ON" or "OFF" of a element or source in a circuit @ $t=0$, will not disturbe the storage element.

$i_L(0^-) \rightarrow$ Just before changing the circuit portion

$i_L(0^+) \rightarrow$ Just After changing the circuit portion

$\boxed{i_L(0^-) = i_L(0^+)}$ storage elements.

In the case of a Capacitor, $\boxed{V_c(0^-) = V_c(0^+)}$

Relationship for Different Parameters

1) Resistor (R)

$V_R(t) = R\,i_R(t)$

or

$i_R(t) = \frac{1}{R} V_R(t)$

Reciprocal of Resistor is denoted by Conductance

$= G V_R(t)$.

2) Inductor (L)	$V_L(t) = L\frac{di_L(t)}{dt}$ $i_L(t) = \frac{1}{L}\int_{-\infty}^{t} V_L(t)\cdot dt$ $= \frac{1}{L}\int_{0}^{t} V_L(t)\cdot dt + i_L(0^+)$ Taking $-\infty \to t$ as initial condition.
3) Capacitor (C)	$V_c(t) = \frac{1}{C}\int_{-\infty}^{t} i_c(t)\cdot dt$ or $i_c(t) = C\frac{dV_c(t)}{dt}$. $V_c(t) = \frac{1}{C}\int_{0}^{t} i_c(t)\cdot dt + V_c(0^+)$

Differential Equations

Type 1: First Order Homogenous differential equations

$$\frac{dy(t)}{dt} + py(t) = 0 \Rightarrow \frac{dy(t)}{y(t)} = -p\cdot dt \Rightarrow \boxed{y(t) = Ke^{-pt}}$$

Type 2: First Order Non-Homogenous differential equations.

$$\frac{dy}{dt}(t) + Py(t) = Q$$

where 'P' is a constant & Q may be a function of independant variable or a constant.

Multipling both sides by "e^{Pt}",

$$e^{Pt}\left[\frac{dy}{dt}(t) + Py(t)\right] = e^{Pt}\cdot Q \Rightarrow \frac{d}{dt}\left[ye^{Pt}\right] = Q\cdot e^{Pt}$$

On Integrating, $ye^{Pt} = \int Q\cdot e^{Pt} + K$,

$$y(t)\cdot e^{Pt} = \int Q e^{Pt} + K \Rightarrow y(t) = e^{-Pt}\int Q\cdot e^{Pt} + Ke^{-Pt}$$

The 1st term of the above Equation is known as the Particular integral which does not contain the arbitary constant, & the 2nd term is known as complementary function, which doesn't depend on the forcing function (input) Qe^{Pt}

$$\boxed{y(t) = \frac{Q}{P} + Ke^{-Pt}}$$

This is the solution for non-homogenous Equation.

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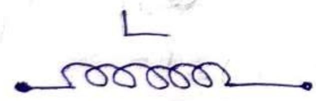
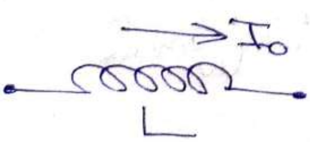
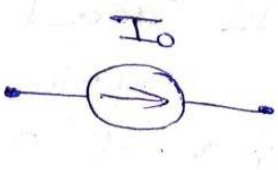
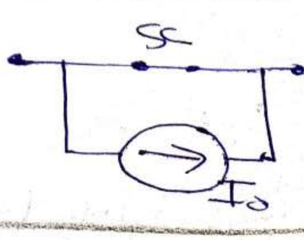
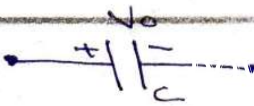
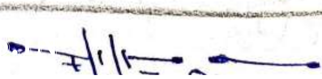
Type III (Second Order Differential Equation)

Let the differential equation be

$$A\frac{d^2y}{dt^2}(t) + B\frac{dy}{dt}(t) + Cy(t) = 0.$$

The general solution is $y(t) = K_1 e^{} + K_2 e^{}$
where K_1 & K_2 are constants
'P_1' and 'P_2' are the roots of the Quadratic equation $AP^2 + BP + C = 0$.
If $P_1 = P_2$, $y(t) = K_1 \cdot e^{P_1 t} + K_2 t \cdot e^{P_1 t}$.

Initial Conditions in Circuits

Element with Initial Condition $t = 0^-$	Equivalent Circuit @ $t = 0^+$	Equivalent Circuit @ $t = \infty$ [Circuit has attain steady state value]
		
		do not allow steady change in current. 
		
		
		

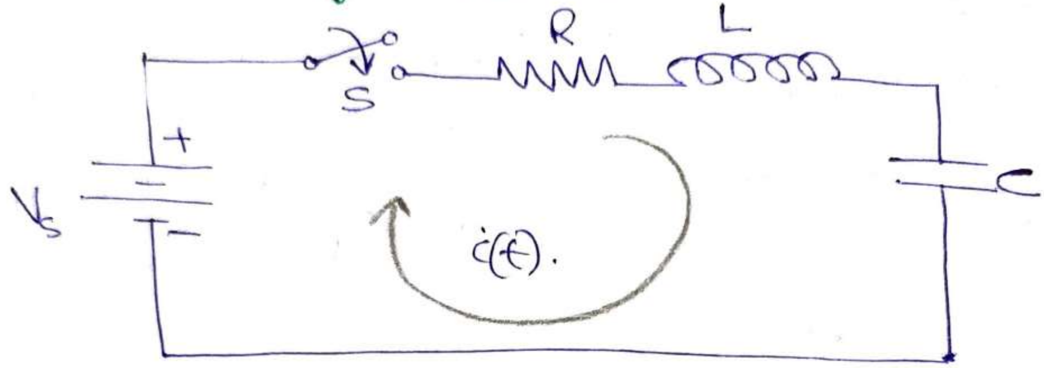
Note: There is no Voltage through the inductor of the Current through it is not changing with time. That is, the inductor is behaving as a short circuit to DC. Similarly, there is no current through the capacitor, if Voltage behaving as an Open circuit to DC.

Different Responses in Dynamic Circuits

1) Transient Response: The Values of Voltage & Current during the transient period are Known as Transient Response. It depends upon the network Elements alone & independent of the forcing function. The complementary function Represents the Transient Response.

2) Steady State Response: The Values of Voltage & Current after the transient has died out are Known as Steady State Response. It depends on both the network Elements & forcing function. The particular integral represents the steady state Response. The complete or total response is the sum of transient Response & steady state Response.

Transcient Response of Series RLC Circuit having DC Excitation



The Values of the parameters are $V_s = 2V$, $R = 6\Omega$, $L = 2H$, $C = 0.25F$.

$$\boxed{R\,i(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)\cdot dt = V_s} \rightarrow (1)$$

Differentiating, we get, $L\frac{d^2i}{dt^2}(t) + R\frac{di}{dt}(t) + \frac{1}{C}i(t) = 0$.

$$2\frac{d^2i}{dt^2}(t) + 6\frac{di}{dt}(t) + \frac{1}{0.25}i(t) = 0$$

$$\Rightarrow \boxed{\frac{d^2i}{dt^2}(t) + 3\frac{di}{dt}(t) + 2i(t) = 0} \rightarrow (2)$$

or $p^2 + 3p + 2 = 0$.

$a = 1,\ b = 3,\ c = 2$

$$p = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-3 \pm \sqrt{9-8}}{2}.$$

$$p = \frac{-3 \pm 1}{2} \Rightarrow \boxed{p = -2, -1}$$

The General Solution of the equation is

$$i(t) = K_1 e^{P_1 t} + K_2 \cdot e^{P_2 t} = K_1 e^{-t} + K_2 \cdot e^{-2t}$$

$$\boxed{i(t) = K_1 e^{-t} + K_2 \cdot e^{-2t}}$$

$i_1(0^+) = \ldots$

The switch is closed at $t=0$,

$i(0^+) = 0$ [Current cannot change instantaneously in an inductor]

substitute the condition in Equation ①

In Equation ①, the 2nd & 3rd voltage term are zero at the instant of switching.

$\boxed{R\,i(0^+) = 0 \quad \because i(0^+) = 0}$

$\frac{1}{C}\int_{-\infty}^{0} i(t)\cdot dt = 0$ [Because the initial voltage across the capacitor is zero.]

$L\frac{di}{dt}(t^{0+}) = V_s$ or $\frac{di}{dt}(0^+) = \frac{V_s}{L} = \frac{2}{2} = 1\,A/sec$

Evaluate the constants 'K_1' & 'K_2', we get

$i(0^+) = 0, \quad \frac{di}{dt}(0^+) = 1$

Now, we can evaluate the constants 'K_1' & 'K_2'.

$i(t) = K_1\cdot e^{-t} + K_2\cdot e^{-2t}$

At, $t = 0^+$, $\boxed{0 = K_1 + K_2}$, $\frac{di(t)}{dt} = -K_1 e^{-t} - 2K_2 e^{-2t}$

At, $t = 0^+$, $\boxed{1 = -K_1 - 2K_2}$, Adding Equations

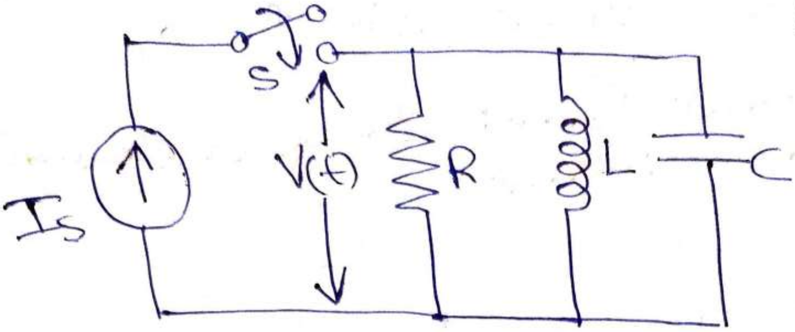
$K_2 - 2K_2 = 1 \Rightarrow \boxed{K_2 = -1,\ K_1 = 1}$

$\boxed{i(t) = e^{-t} - e^{-2t}\,A}$

Homework: Find the response 'V_t' for a parallel RLC circuit.

$I_s = 2A$, $R = \frac{1}{10}\Omega$, $L = \frac{1}{16}H$, $C = 4F$.

Applying KCL same procedure

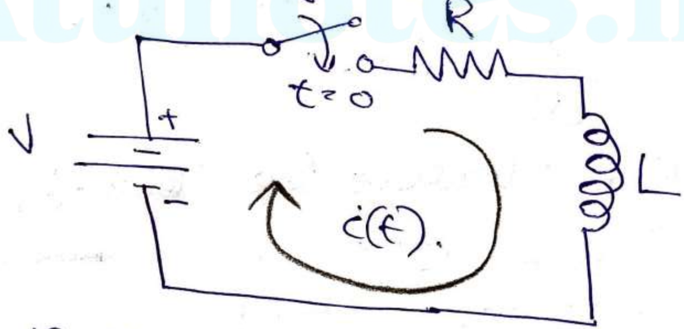


24-9-2020
Thursday

Transient Response of Series RL circuit having DC Excitation

Natural or Source-Free Response of RL circuit

Consider a series RL circuit connected across a DC voltage of 'V' Volt.



Applying KVL in circuit, we get.

$$\boxed{L\frac{di(t)}{dt} + R\,i(t) = V}$$

Rearrange this equation, $\dfrac{di(t)}{dt} + \dfrac{R}{L}i(t) = \dfrac{V}{L}$

Example for non-homogeneous first order differential equation

General Solution: $\boxed{i(t) = \dfrac{V}{R} + Ke^{-\frac{R}{L}t}} \rightarrow (1)$

As we know, the switch is closed @ $t=0$ before that the circuit is open circuited. Therefore the initial current through the inductor is zero.

Assume, $i(0^+) = 0$, sub this in eq (1).

$$0 = \frac{V}{R} + K \Rightarrow \boxed{K = \frac{-V}{R}}$$

$$i(t) = \frac{V}{R} - \frac{V}{R}e^{-R/L\,t} \Rightarrow \boxed{i(t) = \frac{V}{R}\left[1 - e^{-\frac{Rt}{L}}\right]}$$

Source Free response of RL circuit

Voltage Across the Resistor, $V_R(t) = R\,i(t)$.

$$\boxed{V_R(t) = V\left[1 - e^{-\frac{Rt}{L}}\right]}$$

Similarily Voltage Across Inductor,

$$V_L(t) = L\frac{di(t)}{dt} = L\frac{d}{dt}\left[\frac{V}{R} - \frac{V}{R}e^{-\frac{Rt}{L}}\right]$$

$$V_L(t) = L\frac{V}{R}\left[0 - e^{-\frac{Rt}{L}} \times -\frac{R}{L}\right]$$

$$\boxed{V_L(t) = Ve^{-\frac{Rt}{L}}}$$

Imp Relations obtained

$$i(t) = \frac{V}{R}\left[1 - e^{-\frac{Rt}{L}}\right]$$

$$V_R(t) = V\left[1 - e^{-\frac{Rt}{L}}\right]$$

$$V_L(t) = Ve^{-\frac{Rt}{L}}$$

Analyzing the condition of circuit @ $t=0$ & $t=\infty$

At $t=0$

$i(0) = \frac{V}{R}[1-0] \Rightarrow i(0) = 0$

$V_R(0) = 0$

$V_L(0) = V$

Similarily, at $t=\infty$

$i(\infty) = V/R$, $V_R(\infty) = V$, $V_L(\infty) = 0$

when $t = R/L = \tau$

$i(t) = \frac{V}{R}\left[1 - e^{-1}\right]$

$$\boxed{i(t) = 0.632\frac{V}{R}}$$

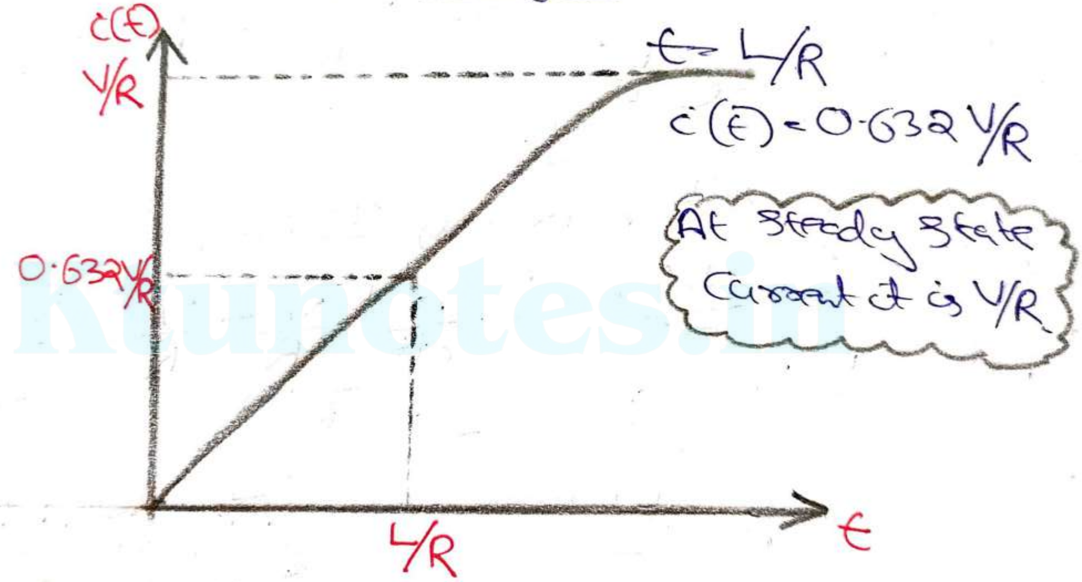
R/L → time constant of series RLC circuit $\tau = R/L$

$t = L/R = \tau \quad , \quad i(t) = \frac{V}{R}\left[1 - e^{-1}\right] = 0.632 \frac{V}{R}$

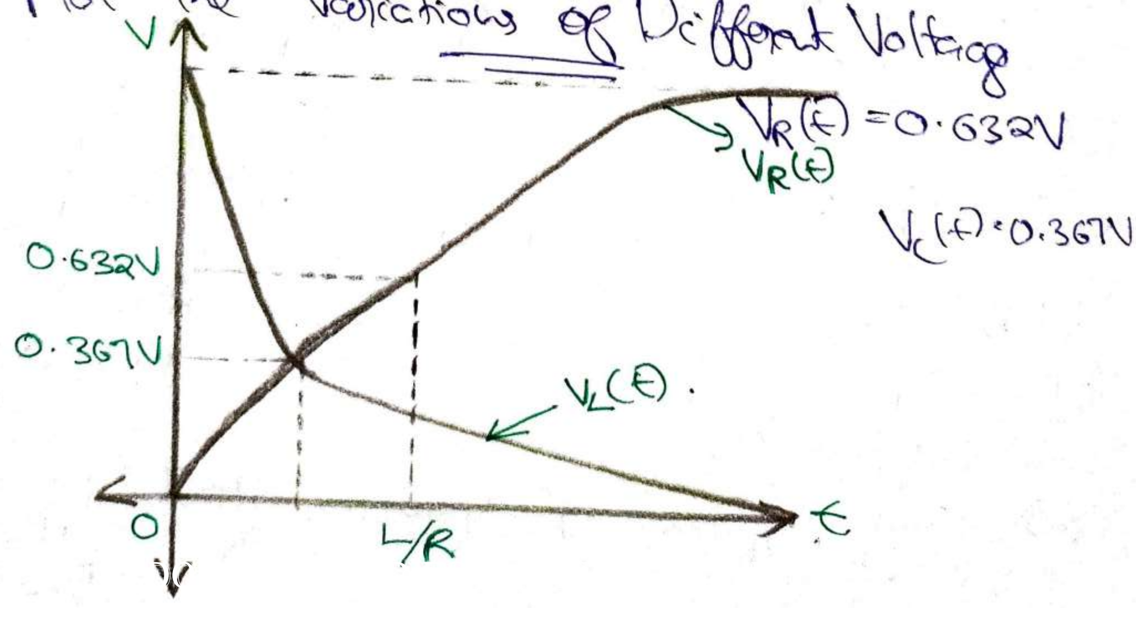
$\boxed{\text{Time Constant } (\tau) = L/R.}$ of the series RL circuit.

Time constant is defined as the interval after which current or voltage change 63.2% of its total change.

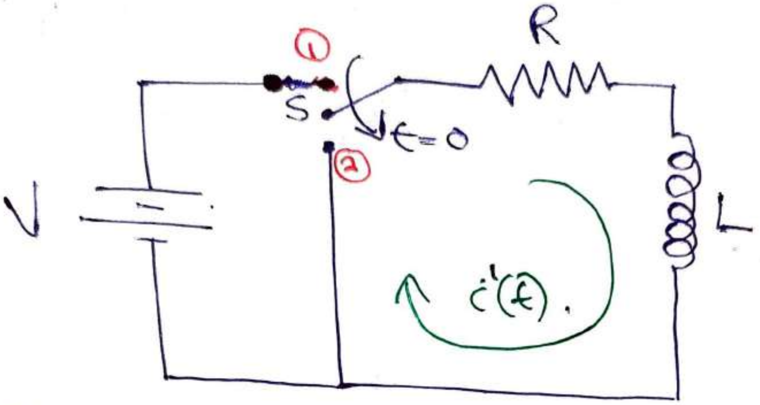
Plot the Response of Series RL Circuit.



Plot the Variations of Different Voltage



Analysis of Another Condition for a Series RL Circuit



Condition: The RL Series Circuit reaches a steady state value & suddenly the voltage is withdrawn by opening the switch and throwing the switch to 2nd position

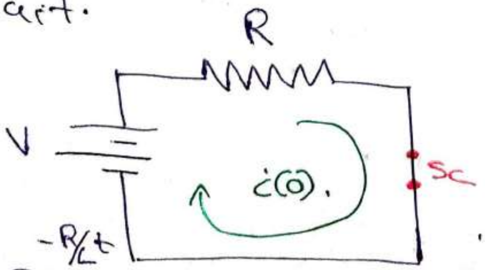
The switch is moved to position 2 at $t=0$.

At position (2), Applying KVL, we get,

$$L\frac{di'(t)}{dt} + Ri'(t) = 0 \Rightarrow \frac{di'(t)}{dt} + \frac{R}{L}i'(t) = 0$$

$$\boxed{i'(t) = Ke^{-R/L\, t}} \longrightarrow (1)$$

At steady state, when $t=\infty$, the inductor acts short circuit.



$i(0^+) = \frac{V}{R}$ Sub initial value condition in equation (1)

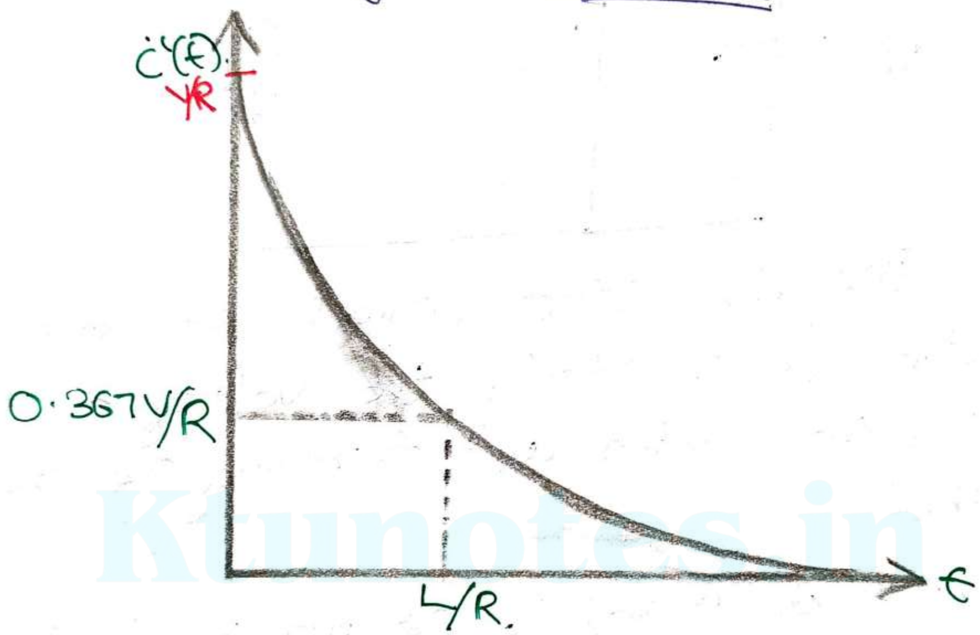
$\frac{V}{R} = K$

$$i'(t) = \frac{V}{R}e^{-R/L\, t}$$

Similarily, the voltage across the Resistor

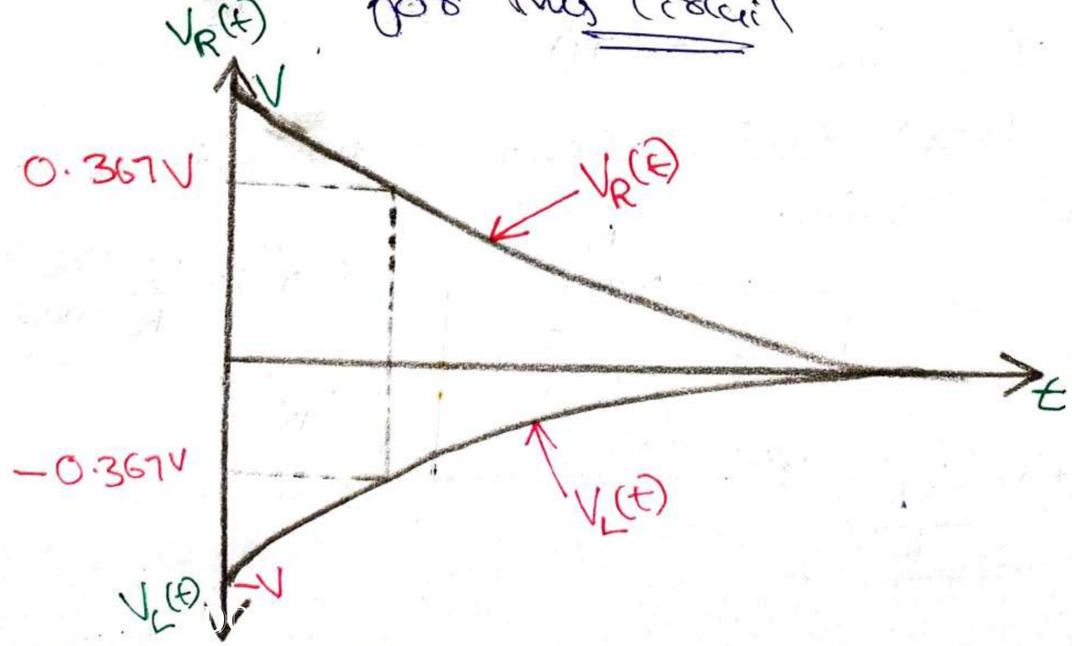
$V_R(t) = Ve^{-R/L\,t}$, $V_L(t) = -Ve^{-\frac{R}{L}t}$

Plot the variation of $i(t)$, $V_R(t)$ & $V_L(t)$ for this Circuit



$t = L/R$, $i(t) = 0.367\, V/R$.

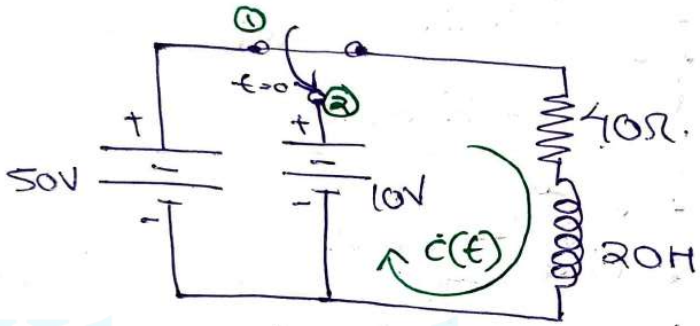
Plot the variation of $V_R(t)$ & $V_L(t)$ for this Circuit



25-9-20 Friday

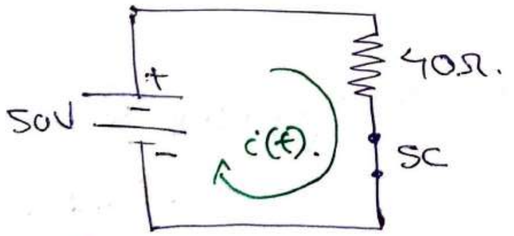
Problem: Analysis of Series RL circuit excited by Constant Voltage or DC Voltage

Q) The Switch in figure is at Position one for a long time and it is moved to position 2 at $t=0$. Obtain the expression for $i(t)$ for $t>0$.



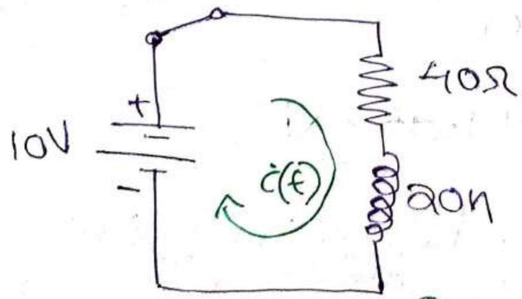
Solution) Given, that the Switch is in position 1 for a long time. The steady state that is attained.

At position 1 for a long time,



$i(0^-) = \frac{50}{40} = \frac{5}{4} A = \underline{1.25A}$.

Therefore, At position 2,



Applying KVL,

$$20\frac{di(t)}{dt} + 40i(t) = 10$$

$$\frac{di(t)}{dt} + 2i(t) = 0.5$$

This is of the form 1st order non-homogenous Differential Equation.

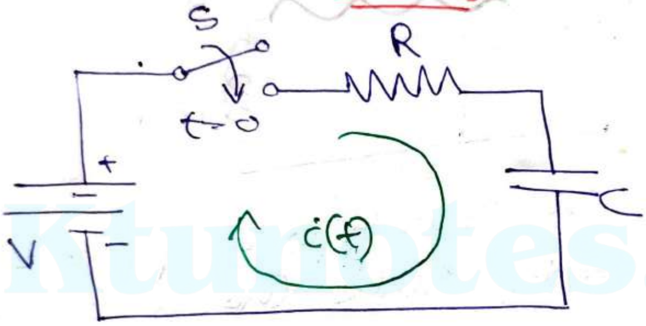
$$i(t) = \frac{0.5}{2} + Ke^{-2t}$$

Sub. the initial Condition, $i(0) = \frac{0.5}{2} + K$

$$1.25 = \frac{0.5}{2} + K \Rightarrow \underline{K = 1}$$

Expression for $i(t) = \underline{0.25 + e^{-2t}}$

Transient Response of Series RC circuit having DC Excitation



Applying KVL,

$$Ri(t) + \frac{1}{C}\int_{-\infty}^{t} i(t)\cdot dt = V$$

$-\infty \to 0$ [initial condition]

$0 \to t$ [transient period]

$$Ri(t) + \frac{1}{C}\int_{0}^{t} i(t)\cdot dt + \boxed{V_c(0^+)} = V$$

→ initial Voltage across the capacitor.

Since the capacitor is initially uncharged

$$V_c(0^+) = 0.$$

$$Ri(t) + \frac{1}{C}\int_{0}^{t} i(t)\cdot dt = V$$

On Differentiating, $R\frac{d}{dt}i(t) + \frac{1}{C}i(t) = 0$

The General Solution is: $i(t) = Ke^{-1/RC \times t}$

At $t = 0^+$, $i(0^+) = V/R$ [∵ the capacitor is short circuited]

Apply this condition to the general solution,

$$\frac{V}{R} = Ke^{0} \Rightarrow K = \frac{V}{R}$$

$$\boxed{i(t) = \frac{V}{R} e^{-\frac{1}{RC}t}}$$

Expressions for $V_R(t)$ & $V_C(t)$

Voltage across the Resistor $V_R(t) = Ri(t) = Ve^{-1/RC \times t}$

Voltage across the Capacitor $V_C(t) = \frac{1}{C}\int_0^{\infty} i(t)\,dt$

$$V_C(t) = V\left[1 - e^{-t/RC}\right] \rightarrow \text{Source-free response}$$

$$V_C(t) = V - V_R(t)$$

Evaluate the circuit condition by $t = 0$ & $t = \infty$

At $t = 0$

$i(t) = \frac{V}{R}$ and $V_R(t) = V$, $V_C(t) = 0$

At $t = \infty$: $i(t) = 0$, $V_R(t)$ & $V_C(t) = V$.

At $t = RC$: $i(t) = \frac{V}{R}e^{-1} = 0.367\frac{V}{R}$

$V_R(t) = 0.367V$, $V_C(t) = 0.632V$

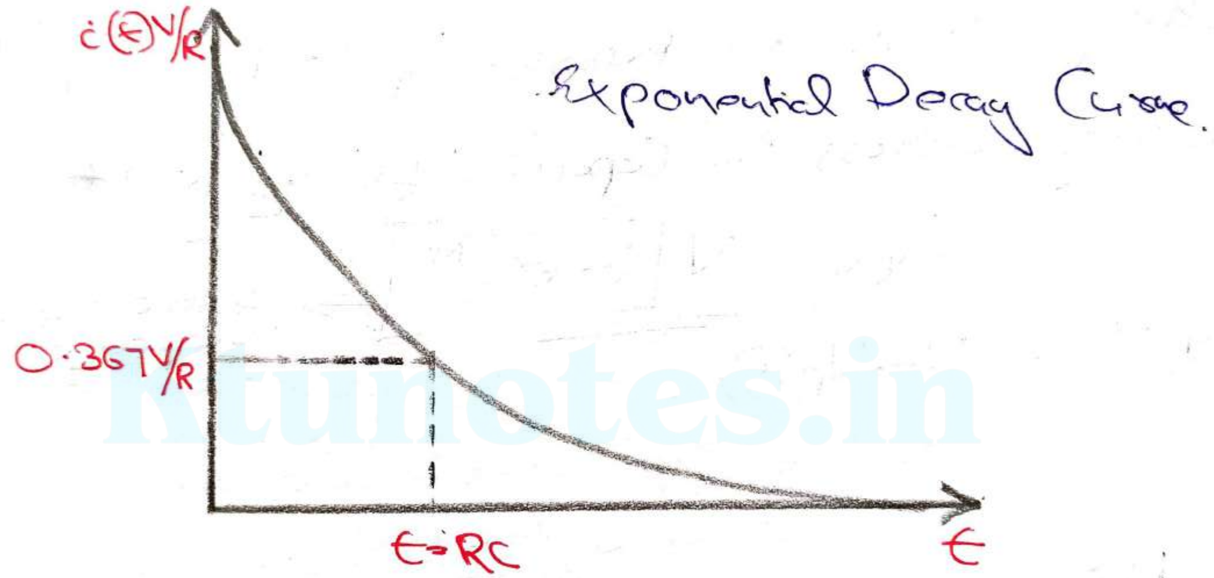
The term "RC" is known as the time constant of a RC circuit. It is denoted (τ).

Time Constant [In Series RC circuit]

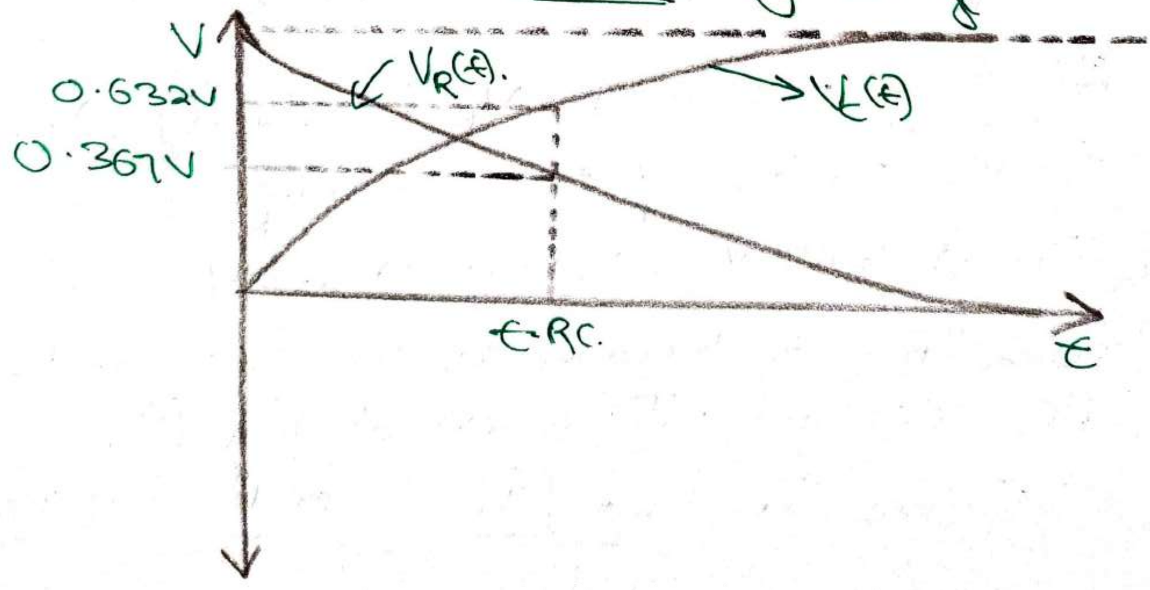
$$\tau = RC$$

And time constant for series RC circuit is the time taken by the response 36.7% of its final ~~value~~ steady state value.

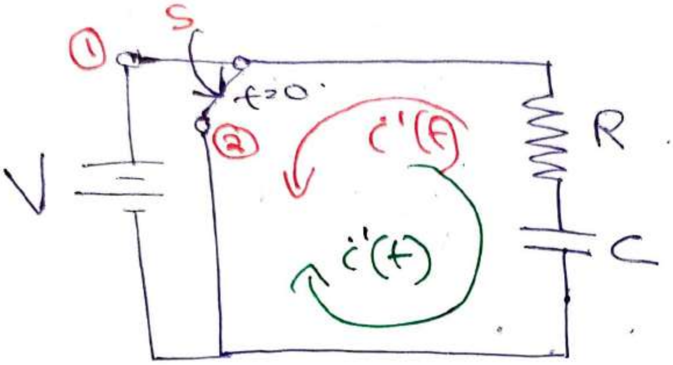
Plot the variation of $i(t)$, $V_R(t)$ and $V_C(t)$



Plot the variation of Voltage



Analysis of Another Condition for a Series RC circuit



At position 2, apply KVL

$$Ri'(t)+\frac{1}{C}\int_{-\infty}^{t} i'(t)\cdot dt = 0$$

$$Ri'(t)+\frac{1}{C}\int_{0}^{t} i'(t)+V_c(0^+) = 0.$$

On Differentiating,
$$R\frac{di'(t)}{dt}+\frac{1}{C}i'(t)=0.$$

$$i'(t) = Ke^{-t/RC}\text{ (for a long time)}$$

At position 1, the switch is closed & the capacitor is fully charged with a value of voltage 'V'. Now, if we switch the position to 2, then the capacitor starts discharging in the opposite direction. Now the current supplied to the circuit, is the voltage in the capacitor in the opposite direction instead of the actual direction.

However, $t=0^+$, the capacitor keeps the steady state voltage $V_c(0^+)=V$ & the direction of $I'(t)$ during discharge is negative. Therefore,

$i'(0^+) = -\frac{V}{R}$. Sub this initial condition,

$\Rightarrow -\frac{V}{R} = Ke^{0} \Rightarrow \boxed{K = -\frac{V}{R}}$

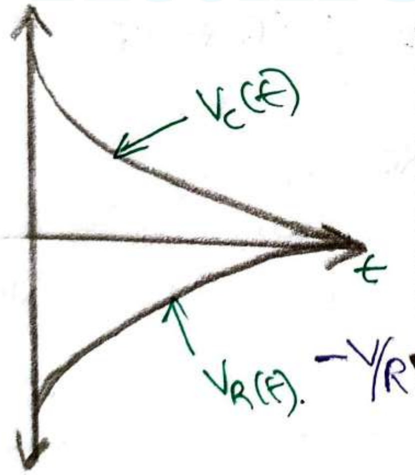
Therefore, General Solution becomes,

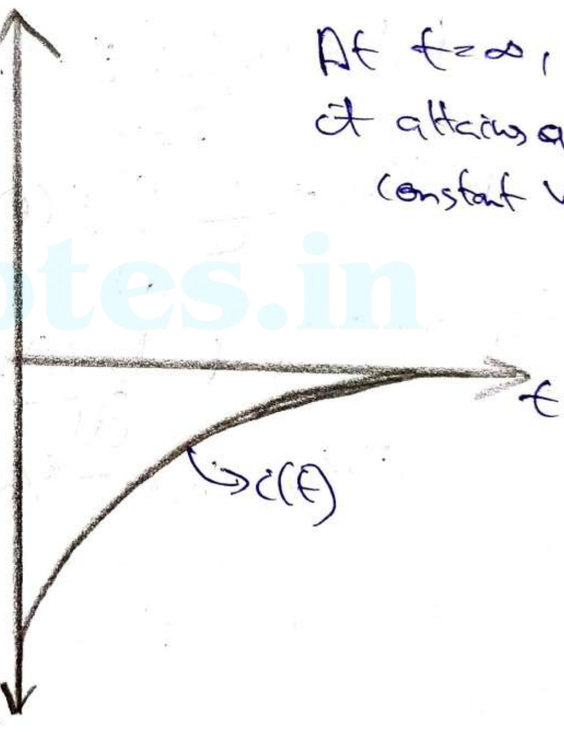
$$\boxed{i'(t) = -\frac{V}{R}e^{-t/RC}}$$

Plot the Variation of $i(t)$

Similarily,

$$\boxed{\begin{aligned} V_R'(t) &= -Ve^{-t/RC} \\ V_C'(t) &= Ve^{-t/RC} \end{aligned}}$$





At $t=\infty$, it attains a constant value

Time Domain Signals $\Longrightarrow$ Frequency Domain Signals

LAPLACE TRANSFORM

$S(t) \longrightarrow F(S)$

30-9-20 Wednesday

Review of Laplace Transforms

It is one of the mathematical tool for the solution of Linear Ordinary integral or differential Equation. The Laplace Transforms converts the integral of differential Equation into a algebraic Equation in 'S' [Laplace's Operator]

Definition of Laplace's Transform: For the time function $f(t)$ which is zero for $t<0$ and that satisfy the condition $\int_0^\infty |f(t)\cdot e^{-\sigma t}|\cdot dt < \infty$ for some real and positive sigma. The Laplace Transform of $f(t)$ is defined as $F(S) = L[f(t)] = \int_0^\infty f(t)\cdot e^{-st}\cdot dt$.

where the variable 's' which is the Laplace's Operator is a Complex Variable, $\boxed{S = \sigma + j\omega}$

Properties of Laplace's Transform

1) Multiplication by a Constant.

$$\boxed{L[Kf(t)] = KF(S)}$$

2) Sum and Differences

$$\boxed{L[f_1(t) \pm f_2(t)] = F_1(S) \pm F_2(S)}$$

3) Differentiation w.r.t. 't'

$$\boxed{L\left[\frac{d}{dt} f(t)\right] = SF(S) - f(0)}$$, where $f(0) = \lim_{t\to 0} f(t)$.

4) Integration by 't'

$$\boxed{L\left[\int_0^t f(t)\cdot dt\right] = \frac{F(s)}{s}}$$

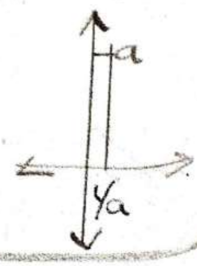
5) Initial Value theorem

$$\boxed{f(0) = \lim_{t\to 0} f(t) = \lim_{s\to\infty} sF(s)}$$

6) Final Value theorem

$$\boxed{f(\infty) = \lim_{t\to\infty} f(t) = \lim_{s\to 0}\left[sF(s)\right]}$$

Laplace Transform Pair

Sl. No.	$f(t)$	$F(s) = \int_0^\infty f(t)\cdot e^{-st}\cdot dt$
1.	1 or Unit Step function $u(t)$, K (constant)	$1/s$, K/s.
2.	t, t^n	$1/s^2$, $\frac{n!}{s^{n+1}}$
3.	$\delta(t)$ [impulse function] (very short duration with infinite magnitude).	Unity (1). 
4.	$e^{\pm at}$,	$\frac{1}{s\mp a}$

5.	$t \cdot e^{\pm at}$	$\frac{1}{(s \mp a)^2}$
6.	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
7.	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
8.	$e^{-at} \cdot \sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
9.	$e^{-at} \cdot \cos \omega t$	$\frac{(s+a)}{(s+a)^2+\omega^2}$
10.	$\sinh(at)$	$\frac{a}{s^2-a^2}$
11.	$\cosh(at)$	$\frac{s}{s^2-a^2}$

* Partial Fraction Expansion Methods (1st method)

1) Partial Fraction Expansion when all the roots are simple

Example: $$F(S) = \frac{P(S)}{(S+S_1)(S+S_2)\cdots(S+S_n)}$$ where the roots $S_1, S_2 \cdots S_n$ are simple

$$F(S) = \frac{K_1}{S+S_1} + \frac{K_2}{S+S_2} + \cdots\cdots + \frac{K_n}{S+S_n}.$$

where, $K_1, K_2 \cdots, K_n$ are residues.

$$K_1 = (S+S_1)F(S) \Big/ S=-S_1$$

General $$K_n = (S+S_n)F(S) \Big/ S=-S_n.$$

Problem Q) Solve the differential equation $x'' + 3x' + 2x = 0$ and $x(0) = 2$ & $x'(0) = -3$

Solution) $$\boxed{L\left[\frac{d^2 f(t)}{dt^2}\right] = S^2F(S) - Sf(0) - f'(0)}.$$

Taking Laplace's Transform,

$$S^2X(S) - Sx(0) - x'(0) + 3\left[SX(S) - x(0)\right] + 2X(S) = 0.$$

$$\Rightarrow S^2X(S) - 2S + 3 + 3SX(S) \underbrace{-3\times 2}_{6} + 2X(S) = 0$$

$$\Rightarrow S^2X(S) + 3SX(S) + 2X(S) = 2S+3$$

$$\Rightarrow (S^2+3S+2)X(S) = 2S+3$$

$$X(S) = \frac{2S+3}{S^2+3S+2} = \frac{2S+3}{(S+1)(S+2)}$$

$$X(S) = \frac{2S+3}{(S+1)(S+2)} = \frac{A}{(S+1)} + \frac{B}{(S+2)}.$$

$$A = (S+1)X(S)\Big|_{S=-1} = \cancel{(S+1)}\frac{(2S+3)}{\cancel{(S+1)}(S+2)}\Big|_{S=-1}$$

$$A = \frac{(2S+3)}{(S+2)}\Big|_{S=-1} \Rightarrow A = \frac{-2+3}{-1+2} \rightarrow \boxed{A=1}$$

$$B = \frac{(2S+3)}{(S+1)}\Big|_{S=-2} \Rightarrow B = \frac{2\times-2+3}{-2+1} = \frac{-1}{-1} = 1$$

$$\boxed{B=1}$$

$$X(S) = \frac{1}{(S+1)} + \frac{1}{(S+2)}$$ taking inverse Laplace transform.

$$x(t) = e^{-t} + e^{-2t}$$

Partial Fraction Expansion when some of the roots are of multiple order (2nd method)

$$F(S) = \frac{P(S)}{(S+S_1)^r (S+S_2)} = \frac{K_{11}}{(S+S_1)} + \frac{K_{12}}{(S+S_1)^2} + \dots + \frac{K_{1r}}{(S+S_1)^r} + \dots + \frac{K_2}{S+S_2}.$$

where $K_{1r} = (S+S_1)^r \cdot F(S)\Big|_{S=-S}$.

$$K_{1(r-1)} = \frac{d}{ds}\left[(S+S_1)^r F(S)\right]_{S=-S_1}$$

Similarly $K_{1(r-2)} = \frac{1}{2!}\frac{d^2}{ds^2}\left[(S+S_1)^r F(S)\right]_{S=-S_1}$

$$\boxed{K_{11} = \frac{1}{(r-1)!}\frac{d^{r-1}}{ds^{r-1}}\left[(S+S_1)^r F(S)\right]\Big|_{S=-S_1}}$$ Important Result.

Problem Q) Find $c(t)$ of $I(S) = \frac{1}{S(S+1)^2(S+2)}$

$$I(S) = \frac{1}{S(S+1)^2(S+2)} = \frac{A}{S} + \frac{B}{(S+2)} + \frac{C_{11}}{(S+1)^2} + \frac{C_{12}}{(S+1)}$$

$A = \left.\frac{1}{(s+1)^2(s+2)}\right|_{s=0} \Rightarrow A = \underline{\underline{\frac{1}{2}}}$

$B = \left.\frac{1}{s(s+1)^2}\right|_{s=-2} = B = \frac{1}{-2(-2+1)^2} = \underline{\underline{\frac{-1}{2}}}$ //

$C_{11} = \left.\frac{1}{s(s+2)}\right|_{s=-1} = C_{11} = \underline{\underline{-1}}$

$\boxed{C_{12} = \frac{d}{ds}\left[\left.(s+1)^2 I(s)\right|_{s=-1}\right] = \frac{d}{ds}\left[\left.\frac{1}{s(s+2)}\right]\right|_{s=-1}} = \underline{\underline{0}}$

Important

$I(s) = \frac{1/2}{s} - \frac{1/2}{s+2} \; \frac{-1}{(s+1)^2}$, Taking Laplace's Inverse,

$\underline{\underline{i(t) = \frac{1}{2} - \frac{1}{2}e^{-2t} - te^{-t} \cdot A}}$

* Partial Fraction Expansion When 2 roots of Complex Conjugate Pair (3rd Method)

if $F(s) = \frac{P(s)}{(s+\alpha+j\omega)(s+\alpha-j\omega)}$

$F(s) = \frac{K_1}{(s+\alpha+j\omega)} + \frac{K_2}{(s+\alpha-j\omega)}$, $K_1 = \left.(s+\alpha+j\omega) \cdot F(s)\right|_{s=-\alpha-j\omega}$

$K_2 = K_1^*$ (conjugate)

Question Q) Find ~~I(t)~~ i(t), if. $I(s) = \frac{s^2+5s+9}{s^3+5s^2+12s+8}$

$I(s) = \frac{s^2+5s+9}{(s+1)(s^2+4s+8)} = \frac{s^2+5s+9}{(s+1)(s+2+j2)(s+2-j2)}$

Put, $s=-1$, $-1+5-12+8=0$,
$(s+1)$

$$\Rightarrow \frac{A}{S+1} + \frac{B}{S+2+j2} + \frac{B^*}{S+2-j2}$$

$$S+1 \overline{\big) S^3+5S^2+12S+8}$$ quotient S^2+4S+8

S^3+S^2

$4S^2+12S$

$4S^2+4S$

$8S+8$

$8S+8$

0

$$\Rightarrow \frac{1}{(S+1)} \; \frac{-1/j4}{(S+2+j2)} + \frac{1/j4}{(S+2-j2)}$$

$$I(S) = \frac{1}{S+1} - \frac{1}{j4(S+2+j2)} + \frac{1}{j4(S+2-j2)}$$

$B = \frac{1}{j-1}$, $B = 1/4$, $B = \frac{1}{-j4}$, $B^* = \frac{+1}{j4}$

Taking Laplace's ~~Trans of~~ Inverse

$$i(t) = e^{-t} - \frac{1}{j4} e^{-(2+j2)t} + \frac{1}{j4} e^{-(2-j2)t}$$

$$= e^{-t} - \frac{1}{j4}\left[e^{-2t}\cdot e^{-j2t} - e^{-2t}\cdot e^{j2t}\right]$$

$$\Rightarrow e^{-t} - \frac{1}{j4} e^{-2t}\left[e^{-j2t} - e^{j2t}\right]$$

$$\Rightarrow e^{-t} + \frac{1}{2} e^{-2t}\left[\frac{e^{j2t} - e^{-j2t}}{j2}\right]$$

$$\boxed{\sin(S) = \frac{e^{iS} - e^{-iS}}{2i} \qquad \cos(S) = \frac{e^{iS} + e^{-iS}}{2}}$$

$$i(t) = e^{-t} + \frac{1}{2} e^{-2t}\cdot \sin 2t \cdot A$$

Another Method

$$I(S) = \frac{S^2+5S+9}{(S+1)(S^2+4S+8)} = \frac{1}{(S+1)} + \frac{1}{S^2+4S+8}$$

$$\frac{A}{S} + \frac{BS+C}{S^2+4S+8} = S^2+5S+9 = A(S^2+4S+8) + (BS+C)S$$

$$I(S) = \frac{1}{S+1} + \frac{1}{S^2+4S+8} = \frac{1}{S+1} + \frac{1}{(S+2)^2+2^2}$$

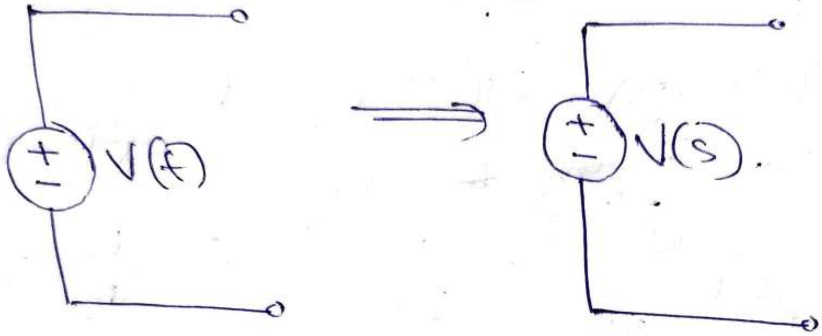
$$I(S) = \frac{1}{S+1} + \frac{1}{2}\cdot\frac{2}{(S+2)^2+2^2}$$

$$i(t) = e^{-t} + \frac{1}{2} e^{-2t}\cdot \sin(2t)\, A.$$

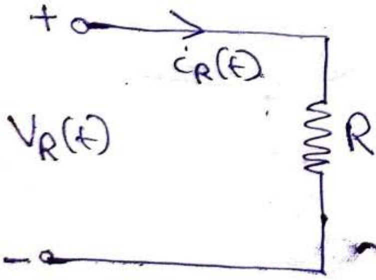
1-10-2020
Thursday

Transformed Circuit Component Representation

Independent Sources



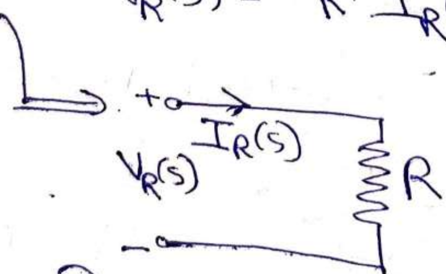
Resistance Parameter



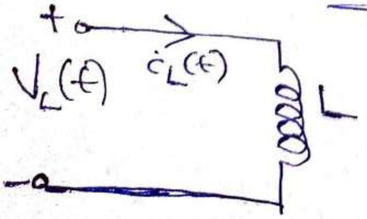
$V_R(t) = R\, i_R(t)$

Changing to into Laplace's Transform

$V_R(s) = R \cdot I_R(s)$.



Inductances Parameter

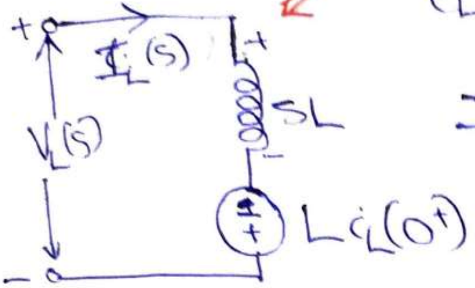


$V_L(t) = L \frac{di_L(t)}{dt}$

$i_L(t) = \frac{1}{L}\int_{0^-}^{t} V_L(t) \cdot dt + i_L(0^+)$

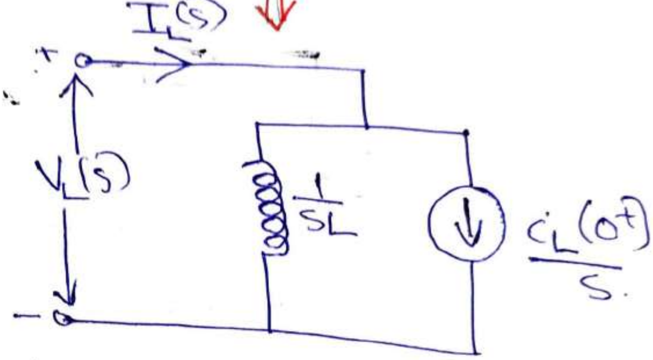
Laplace's Transform $\Rightarrow V_L(s) = L\left[sI_L(s) - i_L(0)\right]$

$$V_L(s) = LsI_L(s) - Li_L(0^+)$$

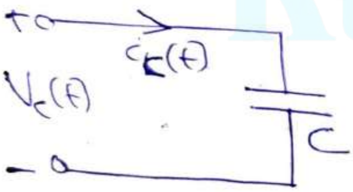


$$i_L(t) = \frac{1}{L}\int_0^t V_L(t)\cdot dt + i_L(0^+)$$

$$I_L(s) = \frac{1}{L}\cdot\frac{V_L(s)}{s} + \frac{i_L(0^+)}{s}$$



Capacitance Parameters

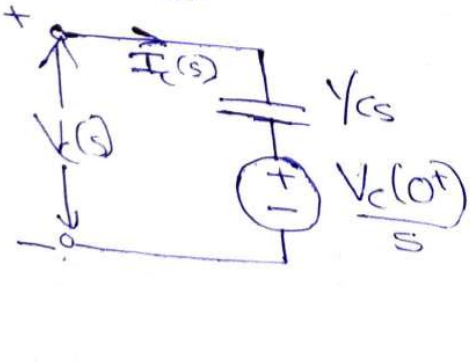


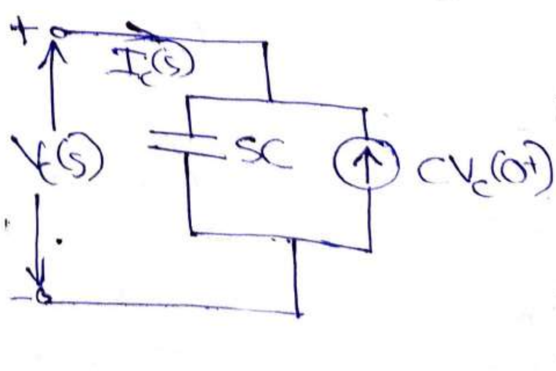
$$V_c(t) = \frac{1}{C}\int_0^t i_c(t) + V_c(0^+)$$

$$i_c(t) = C\frac{dV_c(t)}{dt}$$

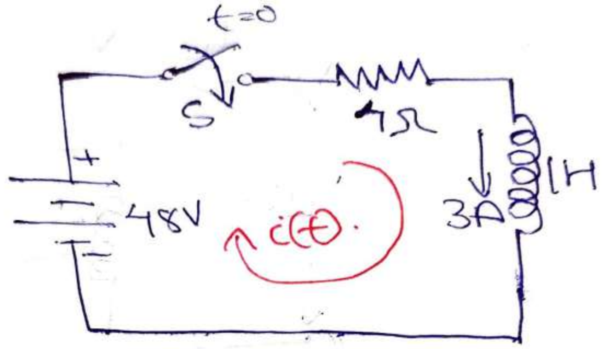
Laplace transform $\Rightarrow V_c(s) = \frac{1}{Cs}I_c(s) + \frac{V_c(0^+)}{s}$

$$i_c(t) = C\frac{dV_c(t)}{dt} \Rightarrow I_c(s) = CsV(s) - CV_c(0^+)$$

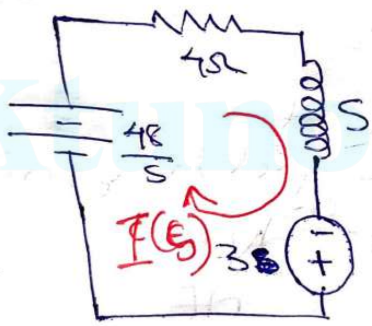




Q) Consider the RL circuit with $R = 4\Omega$ & $L = 1H$ excited by 48 Volt DC Source as shown in Fig. Assume the initial current through inductor as 3A. Using the Laplace's transform. Determine the current $I(t)$. Also Draw the transform circuit.



Solution)



$$4I(s) + SI(s) - 3 = \frac{48}{S}$$

$$I(s)[S+4] = \frac{48+3S}{S}$$

$$I(s) = \frac{3S+48}{S(S+4)}$$

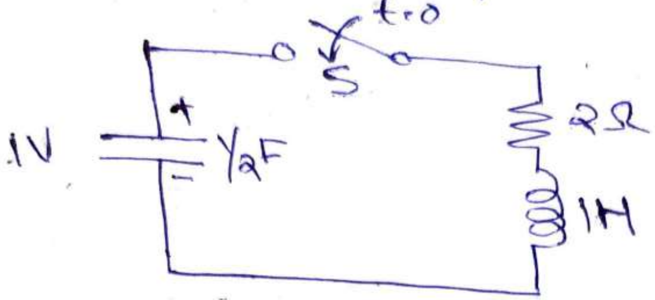
$$I(s) = \frac{A}{S} + \frac{B}{S+4}$$

$$A = \left.\frac{3S+48}{S+4}\right|_{S=0} = A = 12$$

$$B = \left.\frac{3S+48}{S}\right|_{S=-4} = B = -9$$

$$I(s) = \frac{12}{S} - \frac{9}{S+4}$$. Taking L^{-1}, $i(t) = 12 - 9e^{-4t} A$

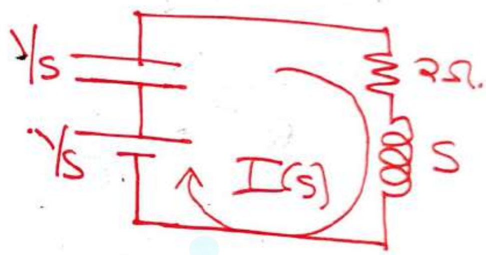
Q) Consider a series RLC circuit with the capacitor initially charged to a Voltage of 1V as shown in Fig. Find an expression for $i(t)$. Also draw the s-domain representation of circuit.



Solution) $L\frac{di(t)}{dt} + Ri(t) + \frac{1}{C}\int_0^t i(t)\,dt + V_c(0^+) = 0$

$$LsI(s) - \cancel{Li(0^+)} + RI(s) + \frac{1}{Cs}I(s) + \frac{V_c(0^+)}{s} = 0$$

$$sI(s) + 2I(s) + \frac{I(s)}{s} + \frac{1}{s} = 0$$



$$I(s)\left\{s + 2 + \frac{1}{s}\right\} = -\frac{1}{s}$$

$$I(s)\left\{\frac{s^2+2s+1}{s}\right\} = -\frac{1}{s}$$

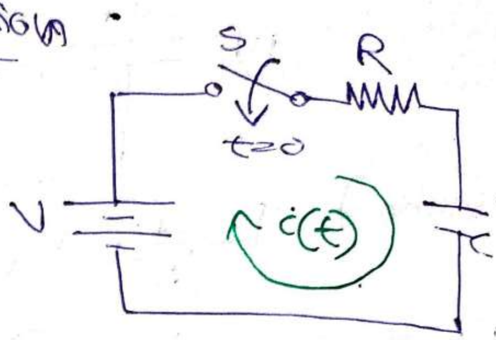
$$I(s) = -\frac{1}{\cancel{s}} \times \frac{\cancel{s}}{s^2+2s+1} \Rightarrow I(s) = -\frac{1}{(s+1)^2}$$

$$i(t) = -t\cdot e^{-t}$$ By Inverse Laplace transform

So, $\boxed{i(t) = -t\cdot e^{-t}\ A}$

2-10-2020 Friday

Response of Series RC circuit with DC excitation.



$$Ri(t)+\frac{1}{C}\int i(t)\cdot dt = V$$

Taking Laplace's Transform

$$RI(s)+\frac{1}{Cs}I(s)=\frac{V}{s}$$

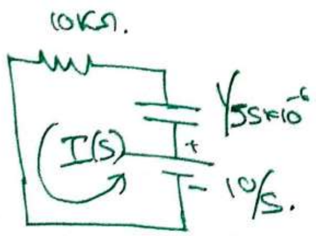
$$I(s)\left[R+\frac{1}{Cs}\right]=V/s \Rightarrow I(s)\left[\frac{RCs+1}{Cs}\right]=V/s.$$

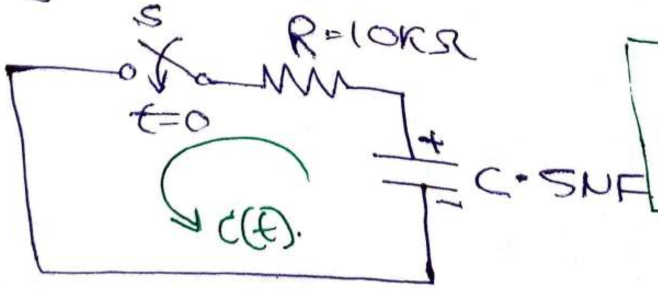
$$I(s)=\frac{CV}{RC\left[s+\frac{1}{RC}\right]}=\frac{V/R}{s+\frac{1}{RC}}$$

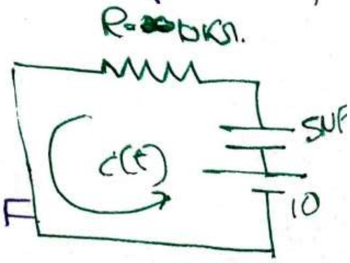
Taking Laplace's Inverse,

$$i(t)=\frac{V}{R}e^{-t/RC}$$

Q) A capacitor of 5μF been charged initially to 10 Volt is connected to a resistance of 10kΩ and is allowed to discharged through it by switching "OFF" a switch 'S'. Find the Expression for discharged current.







$$Ri(t)+\frac{1}{C}\int_{-\infty}^{t} i(t)\cdot dt = 0 \Rightarrow Ri(t)+\frac{1}{C}\left[\int_{-\infty}^{0} i(t)\cdot dt + \right.$$

$\int_0^t i(t)\cdot dt \Big] = 0 \Rightarrow Ri(t) + 10V + \frac{1}{C}\int_0^t i(t)\cdot dt = 0$

Substituting the values, Taking Laplace's transform

$RI(s) + \frac{10}{s} + \frac{1}{Cs} I(s) = 0$

$I(s)\left[R + \frac{1}{Cs}\right] = -\frac{10}{s} \Rightarrow I(s)\left[RCs+1\right] = -10C$

$I(s)\left[s + \frac{1}{RC}\right] = -\frac{10}{R}$ or $I(s) = \dfrac{-\frac{10}{R}}{s + \frac{1}{RC}}$

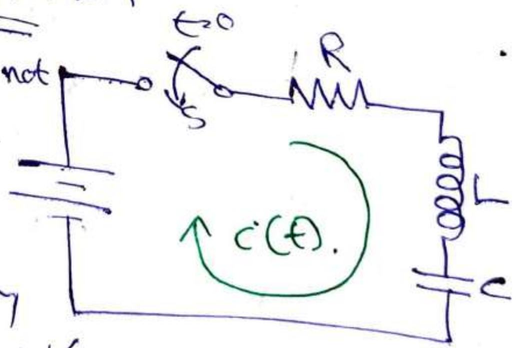
Taking Laplace's Inverse of the equation, we get

$i(t) = -\frac{10}{R} e^{-t/RC} \Rightarrow i(t) = \frac{-10}{10\times10^3} \cdot e^{-t/10\times10^3 \times 5\times10^{-6}}$

$i(t) = -10^{-3} \cdot e^{\frac{-t}{50\times10^{-3}}} \Rightarrow i(t) = -10^{-3} \cdot e^{-20t}$ A

Response of Series RLC circuit with DC excitation

Assumption: No initially stored energy is not present



$Ri(t) + L\frac{di}{dt} + \frac{1}{C}\int i(t)\cdot dt = V$

Taking Laplace's Transform

$RI(s) + LsI(s) + \frac{1}{C} I(s) = V/s$

$\left[R + Ls + \frac{1}{Cs}\right] I(s) = \frac{V}{s} \Rightarrow \left[LCs^2 + RCs + 1\right] I(s) = CV$

$\left[s^2 + \frac{R}{L}s + \frac{1}{LC}\right] I(s) = \frac{V}{L}$

$I(s) = \dfrac{V/L}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$, $s_1 s_2 = \dfrac{-\frac{R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 - 4/LC}}{2}$

$$s_1, s_2 = \frac{-R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

$$s_1 = \frac{-R}{2L} + \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}, \quad s_2 = \frac{-R}{2L} - \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}$$

For simplification, we can make some assumptions, let $\omega_n = \frac{1}{\sqrt{LC}}$ (5th sem (control system))

$\omega_n \rightarrow$ natural frequency of oscillation (rad/sec)

$\delta\omega_n = \frac{R}{2L}$, where $\delta \rightarrow$ damping ratio

$$s_1 = -\delta\omega_n + \sqrt{(\delta\omega_n)^2 - \omega_n^2} = -\delta\omega_n + \omega_n\sqrt{\delta^2 - 1}$$

$$s_2 = -\delta\omega_n - \sqrt{(\delta\omega_n)^2 - \omega_n^2} = -\delta\omega_n - \omega_n\sqrt{\delta^2 - 1}$$

$$I(s) = \frac{V/L}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{V/L}{(s - s_1)(s - s_2)}$$

Apply partial fraction,

$$I(s) = \frac{A}{s - s_1} + \frac{B}{s - s_2}, \text{ where } A = \frac{V/L}{s_1 - s_2}, \quad B = \frac{V/L}{s_2 - s_1}$$

$$A = \frac{V/L}{2\omega_n\sqrt{\delta^2 - 1}}, \quad B = \frac{-V/L}{2\omega_n\sqrt{\delta^2 - 1}}$$

$$I(s) = \frac{V/L}{2\omega_n\sqrt{\delta^2 - 1}}\left[\frac{1}{s - s_1} - \frac{1}{s - s_2}\right]$$

Taking Inverse Laplace Transform

$$i(t) = \frac{V/L}{2\omega_n\sqrt{\delta^2 - 1}}\left[e^{s_1 t} - e^{s_2 t}\right]$$

$$i(t) = \frac{V/L}{2\omega_n\sqrt{\delta^2-1}}\left[e^{[-\delta\omega_n+\omega_n\sqrt{\delta^2-1}]t} - e^{[-\delta\omega_n-\omega_n\sqrt{\delta^2-1}]t}\right]$$

$$i(t) = \frac{V/L}{2\omega_n\sqrt{\delta^2-1}} \cdot e^{-\delta\omega_n t}\left[e^{(\omega_n\sqrt{\delta^2-1})t} - e^{-(\omega_n\sqrt{\delta^2-1})t}\right]$$

There are 3 cases for 'δ'

1) $\delta > 1$, 2) $\delta = 1$, 3) $\delta < 1$

$\boxed{\text{undamped } \delta = 0}$

$\boxed{\text{Case 1: } \delta > 1}$

$\boxed{\sin(hat) = \frac{e^{at} - e^{-at}}{2}}$

$\left(\frac{R}{2L}\right)^2 > \frac{1}{LC}$, $Q = \omega_0 L/R$

$$i(t) = \frac{V}{\omega_n L\sqrt{\delta^2-1}} \cdot e^{-\delta\omega_n t} \sinh(\omega_n\sqrt{\delta^2-1}) \cdot t$$

This condition is said to be Over Damped

$\boxed{\text{2) } \delta = 1 : \text{Case 2:}}$ [Critically damped]

$\left(\frac{R}{2L}\right)^2 = \frac{1}{LC}$, $s_1 = -\omega_n$, $s_2 = -\omega_n$.

$I(s) = \frac{V/L}{(s+\omega_n)^2}$, $i(t) = \frac{V}{L} t \, e^{-\omega_n t}$

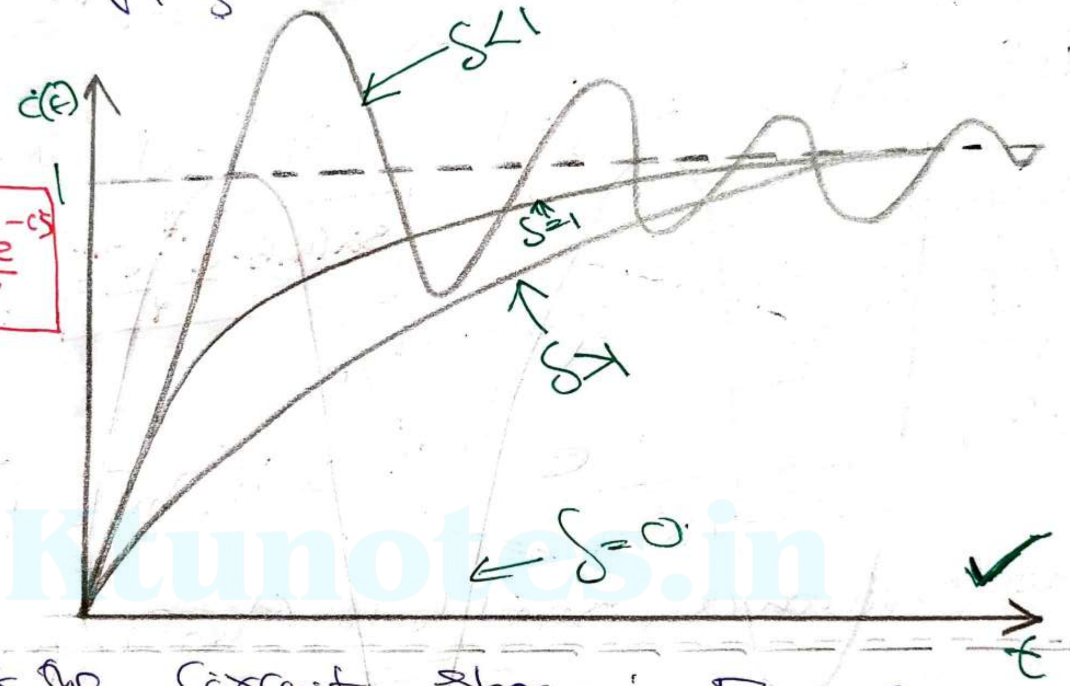
3) $\boxed{\text{Case 3: } \delta < 1}$ [Under Damped]

$\left(\frac{R}{2L}\right)^2 < \frac{1}{LC}$, $i(t) = \frac{V/L}{\omega_n\sqrt{1-\delta^2}}\left[e^{-\delta\omega_n t}\left[\frac{e^{j\omega_n\sqrt{1-\delta^2}t} - e^{-j\omega_n\sqrt{1-\delta^2}t}}{2j}\right]\right]$

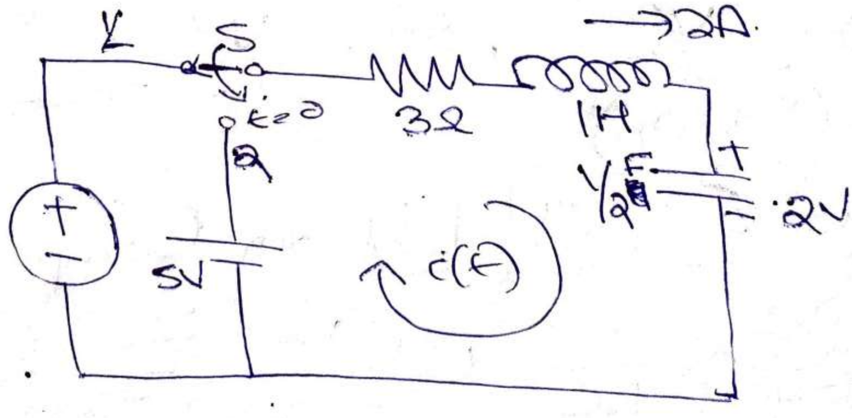
$$i(t) = \frac{\frac{V}{L}}{\omega_n\sqrt{1-\delta^2}} \cdot e^{-\delta\omega_n t}\left[\frac{e^{j\omega_n\sqrt{1-\delta^2}t} - e^{-j\omega_n\sqrt{1-\delta^2}t}}{2j}\right]$$

$$i(t) = \frac{V}{\omega_n L\sqrt{1-\delta^2}} \cdot e^{-\delta\omega_n t} \sin \underline{\omega_n\sqrt{1-\delta^2}t}$$

$$\boxed{\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}}$$



Q) For the circuit shown in Fig. the switch is moved from position-1 to position-2 at t=0. Find i(t), if $I_L(0^-)$ is 2A & $V_C(0^-)$ is 2V.



With Switch in position 1,

$i_L(0^-) = 2A, \quad V_c(0^-) = 2V.$

With Switch in position 2,

$$Ri(t) + L\frac{di}{dt} + \frac{1}{C}\int_{-\infty}^{t} i(t)\cdot dt = 5$$

$$Ri(t) + L\frac{di}{dt} + \frac{1}{C}\int_{0}^{t} i(t)\cdot dt + 2 = 5$$

Taking Laplace's transform of the Equation,

$$3I(s) + S[\cancel{I(s)}\; SI(s) - \underset{\uparrow 2A}{i(0^-)}] + \frac{2I(s)}{S} + \frac{2}{S} = \frac{5}{S}$$

$$3I(s) + SI(s) - 2 + \frac{2}{S}I(s) = \frac{3}{S}$$

$$I(s)\left[3 + s + \frac{2}{S}\right] = \frac{3}{S} + 2$$

$$I(s) = \frac{\frac{3}{S}+2}{3+s+\frac{2}{S}} \Rightarrow \frac{2s+3}{(s+1)(s+2)}$$

~~i(t) = f(2s+3)~~ $\quad i(t) = e^{-t} + e^{-2t}\ A$

By Partial Fraction Expansion,

$$\frac{2s+3}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$A = \left.\frac{2s+3}{s+2}\right|_{s=-1} \Rightarrow A = \frac{-2+3}{-1+2} = 1$$

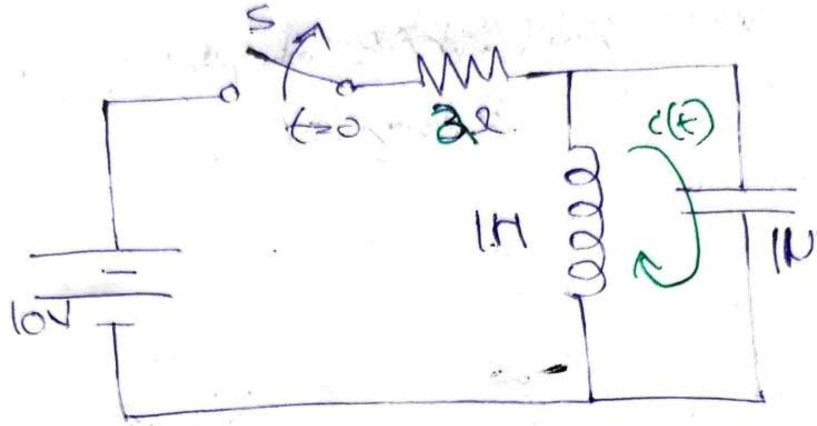
$$B = \left.\frac{2s+3}{s+1}\right|_{s=-2} \Rightarrow B = \frac{-4+3}{-2+1} = 1$$

$$I(s) = \frac{1}{s+1} + \frac{1}{s+2}$$

Taking Laplace Inverse

$$\boxed{i(t) = e^{-t} + e^{-2t}\ A}$$

Q) For the circuit shown in Figure, the Switch 'S' is closed, and steady state condition is reached, at t=0, the switch is open. Obtain the Expression for current through the Inductor. $\frac{12\,Q_2}{CA}$

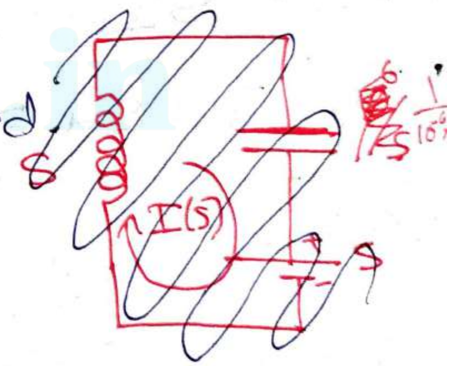


There is no voltage across the capacitor since inductor is short circuited.

At steady state, $i_L(0^-) = \frac{10}{2} = \underline{\underline{5A}}$

$V_C(0^-) = \underline{\underline{0}}$ [Since inductor is short circuited]

At $t=0$, switch is opened.



Apply KVL in circuit

$$\frac{di}{dt} + \frac{1}{10^{-6}}\int i(t)\cdot dt = 0$$

$$sI(s) - i_L(0^-) + \frac{1}{s\times 10^{-6}} I(s) = 0$$

$$s\cdot I(s) - 5 + \frac{10^6}{s} I(s) = 0$$

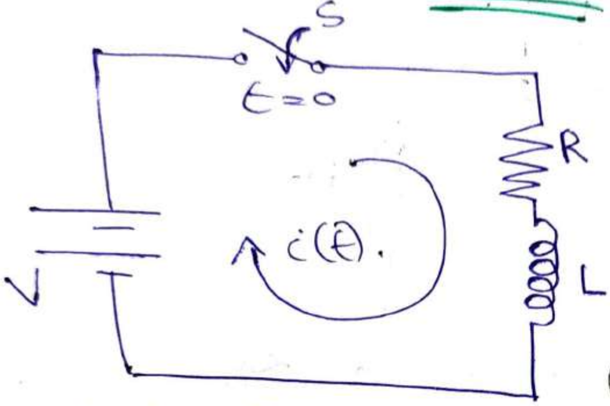
$$I(s)\cdot\left[s + \frac{10^6}{s}\right] = 5 \Rightarrow I(s) = \underline{\underline{\frac{5s}{s^2+10^6}}} \Rightarrow \frac{5s}{s^2+(10^3)^2}$$

$$i(t) = \underline{\underline{5\cos(1000)t \text{ A}}}$$

2-10-2021 Friday

~~Transformed Circuit Component Representation~~

Step - Response of Series RL Circuit



Let us assume that inductance is initially unenergized (That is there is no initially current flowing through it)

Writing KVL Equation,

$Ri(t) + L\frac{di}{dt}(t) = V$, Taking Laplace Transform,

$RI(s) + LSI(s) - \cancel{Li(0^+)} = V/S$

↳ initial current through inductor.

Since, we assume that, there is no initial current in the inductor.

$I(s)\left[SL + R\right] = V/S \Rightarrow I(S)L\left[S + \frac{R}{L}\right] = V/S$

$I(S) = \frac{V}{L}\,\frac{1}{S(S+\frac{R}{L})}$

Now, we can apply Partial Fraction Technique,

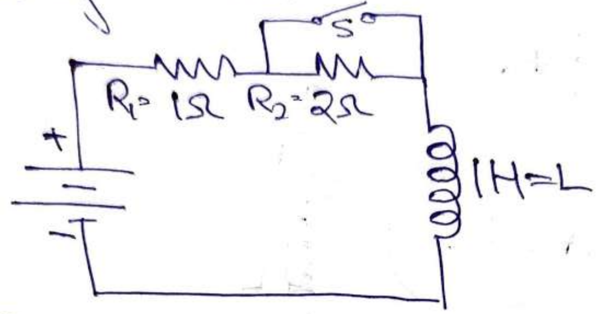
$\cancel{I(s)} = \frac{V/L}{S(S+R/L)} = \frac{A}{S} + \frac{B}{S+\frac{R}{L}}$

$A = \left.\frac{V/L}{S+R/L}\right|_{S=0} = \frac{V}{R}$, $B = \left.\frac{V/L}{S}\right|_{S=-R/L} = \frac{-V}{R}$

$I(S) = \frac{V/R}{S} - \frac{V/R}{S+R/L}$ → Taking Laplace's Inverse,

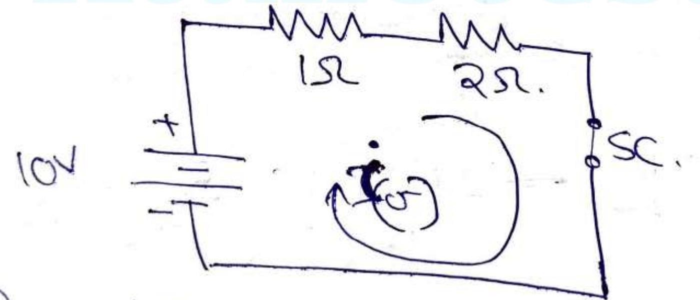
$i(t) = V/R - V/R\, e^{-R/L\,t} \Rightarrow \frac{V}{R}\left[1 - e^{-R/L\,t}\right]$ → Forced Response

Q) In the circuit shown in Figure, the Battery Voltage is applied for a steady state period. Obtain the complete expression for the current after closing the switch 'S'.



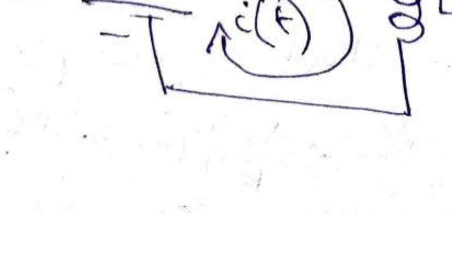
Solution) The circuit is at steady state condition before closing the switch.

Before closing the s/w, the circuit attains steady state & inductor is short circuited.



$$i(0^-) = \frac{10}{3} A$$

At t=0, the switch is closed, the R_2 becomes short-circuited ∴ we get, the new circuit.



$$Ri(t) + L\frac{di}{dt} = V$$

Taking Laplace's Transform

$$RI(s) + L\left[sI(s) - i(0^+)\right] = V/s$$

$i(0^+) = i(0^-) = \frac{10}{3}A \quad RI(s) + LsI(s) - Li(0^+) = V/s$

$I(s)\left[sL+R\right] = V/s - Li(0^+)$

$I(s)\left[s+1\right] = \frac{10}{s} - \frac{10}{3} \Rightarrow I(s) = \frac{10}{s(s+1)} - \frac{10/3}{s+1}$

$\frac{10}{s(s+1)} = \frac{A}{s} + \frac{B}{s+1} \Rightarrow A = \frac{10}{s+1}\Big|_{s=0} = 10.$

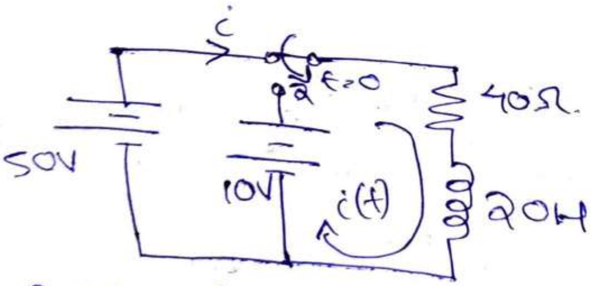
$B = \frac{10}{s}\Big|_{s=-1} = \underline{\underline{-10}}$

$I(s) = \frac{10}{s} - \frac{10}{s+1} - \frac{10/3}{s+1} = \frac{10}{s} - \frac{6.67}{s+1}$

Taking Laplace's Inverse,

$i(t) = 10 - \underline{\underline{6.67e^{-t}A}}$

Q) For the RL series circuit, shown in Fig, the switch is position 1 for a long time, & then moved to position 2 at $t=0$. Obtain the expression for $i(t)$. for $t>0$



Solution) With the switch in position (1), the initial current $i(0^-) = \frac{50}{40} = \underline{\underline{1.25A}}$

[Inductor is behaving as a short circuit]

With the switch in position (2),

Apply KVL, $40i(t) + 20\frac{di}{dt} = 10$

$40I(s) + 20\left[sI(s) - i(0^+)\right] = 10/s$

$$40I(s) + 20sI(s) - 20\times1.25 = 10/s$$

$$I(s)\left[20s+40\right] = \frac{10}{s} + 25$$

$$I(s) = \frac{\frac{10}{s}+25}{20s+40} = \frac{10}{20s[s+2]} + \frac{25}{20(s+2)}$$

$$I(s) = \frac{0.5}{s(s+2)} + \frac{1.25}{s+2}$$

$$\frac{0.5}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} \Rightarrow A = \left.\frac{0.5}{s+2}\right|_{s=0} = \frac{0.5}{2} = \underline{\underline{0.25}}$$

$$B = \left.\frac{0.5}{s}\right|_{s=-2} = \underline{\underline{-0.25}}$$

$$I(s) = \frac{0.25}{s} - \frac{0.25}{s+2} + \frac{1.25}{s+2}$$

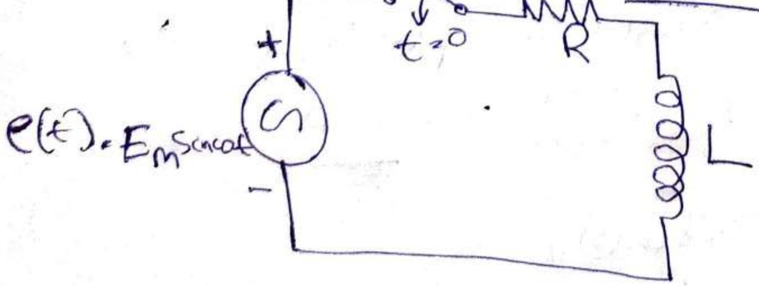
Taking Inverse Laplace,

$$i(t) = 0.25 - 0.25e^{-2t} + 1.25e^{-2t}$$

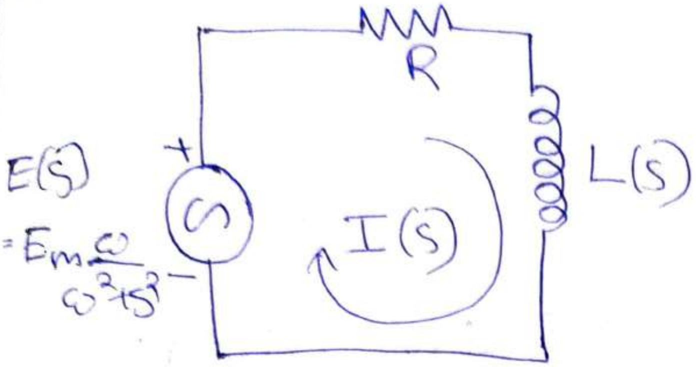
$$i(t) = \underline{\underline{0.25 + e^{-2t}}}\ A.$$

7-10-2020
Wednesday

Transient Analysis of Series RL Circuit excited by AC Source



There is no initial stored energy in the circuit.



$$I(s) = \frac{E(s)}{R+Ls}$$

$$= \frac{E_m \frac{\omega}{\omega^2+s^2}}{L\left[s+\frac{R}{L}\right]}$$

$$\Rightarrow \frac{\omega E_m}{L} \times \frac{\omega E_m/L}{\left(s+\frac{R}{L}\right)(s^2+\omega^2)}$$ Applying Partial fraction method,

$$\Rightarrow \frac{A}{s+\frac{R}{L}} + \frac{Bs+C}{s^2+\omega^2}$$

Where $A = \left.\frac{\omega\ E_m/L}{s^2+\omega^2}\right|_{s=-\frac{R}{L}}$ $\Rightarrow A = \frac{\frac{\omega E_m}{L}}{\frac{R^2}{L^2}+\omega^2}$

$$A = \frac{\omega E_m}{\cancel{L}} \times \frac{L^2}{R^2+\omega^2 L^2} \Rightarrow A = \frac{\omega E_m L}{R^2+\omega^2 L^2} \Rightarrow \frac{\omega L\cdot E_m}{R^2+\omega^2 L^2}$$

$$Z = \sqrt{R^2+X_C^2} = \sqrt{R^2+\omega^2 L^2}$$

$$Z^2 = R^2+\omega^2 L^2 \qquad A = \frac{\omega L\cdot E_m}{Z^2}$$

$$\frac{\omega E_m}{L} = A(s^2+\omega^2)+(Bs+C)\left(s+\frac{R}{L}\right)$$

$$= As^2 + A\omega^2 + Bs^2 + \frac{BsR}{L} + Cs + \frac{CR}{L}$$

Equating Co-efficients of s^2,

$$A+B = 0 \text{ or } B = -A = -\frac{\omega L E_m}{Z^2}$$

$$\frac{BR}{L}+C = 0 \Rightarrow C = -\frac{BR}{L}$$

$$B = \frac{\omega \cancel{L} E_m R}{\cancel{L}\cdot Z^2} \Rightarrow B = \frac{\omega E_m R}{Z^2}$$

$$I(s) = \frac{\frac{\omega L E_m}{Z^2}}{s+\frac{R}{L}} + \frac{-\frac{\omega L E_m}{Z^2}\cdot s + \frac{\omega R E_m}{Z^2}}{s^2+\omega^2}$$

$$I(s) = \frac{\omega L E_m}{Z^2}\cdot\frac{1}{s+\frac{R}{L}} - \frac{\omega L E_m}{Z^2}\times\frac{s}{s^2+\omega^2} + \frac{R E_m}{Z^2}\cdot\frac{\omega}{s^2+\omega^2}$$

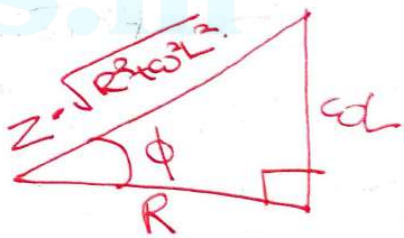
$$i(t) = \frac{\omega L E_m}{Z^2}\cdot e^{-\frac{R}{L}t} - \frac{\omega L E_m}{Z^2}\cdot\cos(\omega t) + \frac{R E_m \sin\omega t}{Z^2}$$

$$i(t) = \frac{\omega L\cdot E_m}{Z^2}\cdot e^{-\frac{R}{L}t} + \frac{E_m}{Z^2}\left[R\cdot\sin\omega t - \omega L\cdot\cos\omega t\right]$$

Construct a Right Angled triangle,

$$\cos\phi = \frac{R}{Z}, \quad \sin\phi = \frac{\omega L}{Z}$$

$$\boxed{R = Z\cdot\cos\phi} \qquad \boxed{\omega L = Z\cdot\sin\phi}$$



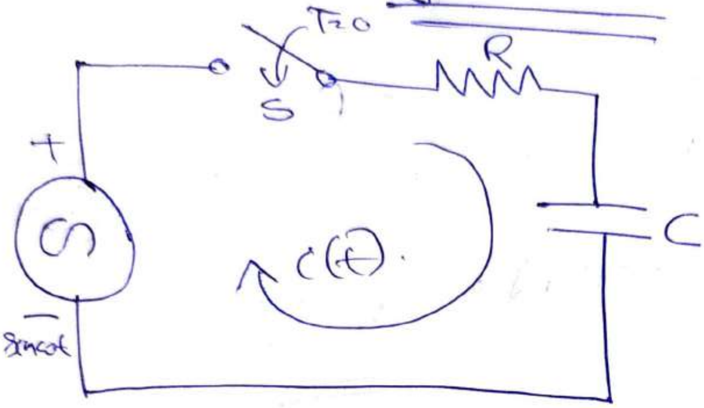
$$\cancel{i(t) = \frac{Z\sin\phi\, E_m}{Z^2}\cdot e}$$

$$i(t) = \frac{\omega L\cdot E_m}{Z^2}\, e^{-\frac{R}{L}t} + \frac{E_m}{Z^2}\left[Z\cdot\cos\phi\cdot\sin\omega t - Z\sin\phi\cdot\cos\omega t\right]$$

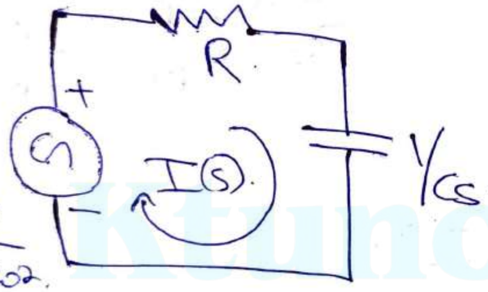
$$i(t) = \frac{\omega L\cdot E_m}{Z^2}\, e^{-\frac{R}{L}t} + \frac{E_m}{Z}\left[\cos\phi\cdot\sin\omega t - \cos\omega t\cdot\sin\phi\right]$$

$$i(t) = \underbrace{\frac{\omega L\cdot E_m}{Z^2}\, e^{-\frac{R}{L}t}}_{\text{Transcient Part}} + \underbrace{\frac{E_m}{Z}\cdot\sin(\omega t-\phi)}_{\text{Steady State Part}}$$

Transcient Analysis of Series RC Circuit excited by AC Source.



$e(t) = E_m \sin\omega t$



$E(s) = E_m \dfrac{\omega}{s^2+\omega^2}$

$$I(s) = \frac{E(s)}{R + \frac{1}{Cs}} \Rightarrow \frac{E(s)}{\frac{R(s+1)}{Cs}}$$

$$\Rightarrow \frac{\cancel{C} s E_m \frac{\omega}{s^2+\omega^2}}{R\cancel{C}\left[s + \frac{1}{RC}\right]} \Rightarrow I(s) = \frac{s E_m \cdot \frac{\omega}{\omega^2+s^2}}{R\left[s + \frac{1}{RC}\right]}$$

$$I(s) = \frac{\omega E_m}{R} \cdot \frac{s}{\left(s+\frac{1}{RC}\right)\left(s^2+\omega^2\right)}$$

Applying Partial Fraction Technique,

$$I(s) = \frac{A}{s+\frac{1}{RC}} + \frac{Bs+C}{s^2+\omega^2}$$

$$A = \frac{\omega \cdot E_m}{R} \cdot \frac{s}{s^2+\omega^2}\Bigg|_{s=-1/RC}$$

$$A = \frac{\omega E_m}{R} \times \frac{-1}{RC(s^2+\omega^2)} \Rightarrow A = \frac{-\omega E_m}{R^2C(s^2+\omega^2)}$$

$$A = \frac{\frac{-E_m}{\omega C}}{R^2+\left(\frac{1}{\omega C}\right)^2} \Rightarrow A = \frac{-E_m}{Z^2\omega C}$$

$$B = \frac{E_m}{Z^2\omega C}, \quad C = \frac{\omega E_m}{R} - \frac{B}{RC} - \frac{\omega E_m}{R} - \frac{E_m}{Z^2\omega C\,RC}$$

$$\Rightarrow \frac{Z^2\omega C^2 \times \omega E_m - E_m}{Z^2\omega C^2 R}$$

$$\Rightarrow E_m\left[\frac{\omega^2C^2Z^2-1}{\omega C^2Z^2R}\right] \Rightarrow E_m\left[\cancel{\omega^2C^2}\right.$$

$$C = \frac{E_m}{\omega RC^2Z^2}\left[\omega^2C^2\left[R^2+\left(\frac{1}{\omega C}\right)^2\right]-1\right]$$

$$C = \frac{E_m}{\omega RC^2Z^2}\left[\omega^2C^2R^2+1-1\right]$$

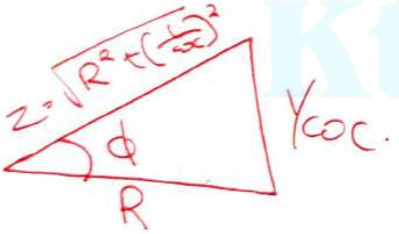
$$C = \frac{E_m\,\omega R}{Z^2}$$

$$I(S) = \frac{-\frac{E_m}{Z^2\omega C}}{S+\frac{1}{RC}} + \frac{\left(\frac{E_m}{Z^2\omega C}\right)s + \frac{E_m\cdot\omega R}{Z^2}}{S^2+\omega^2}$$

$$I(S) = \frac{-E_m}{Z^2\omega C}\cdot\frac{1}{S+\frac{1}{RC}} + \frac{E_m}{\omega C Z^2}\cdot\frac{1}{S^2+\omega^2} + \frac{RE_m}{Z^2}\cdot\frac{\omega}{S^2+\omega^2}$$

$$i(t) = \frac{-E_m}{\omega C Z^2}\cdot e^{-t/RC} + \frac{E_m}{\omega C Z^2}\cdot\cos(\omega t) + \frac{RE_m}{Z^2}\cdot\sin(\omega t)$$

$$i(t) = \frac{-E_m}{\omega C Z^2}\cdot e^{-\frac{1}{RC}t} + \frac{E_m}{Z^2}\left[\cos(\omega t)\cdot\frac{1}{\omega C} + \sin(\omega t)\cdot R\right]$$

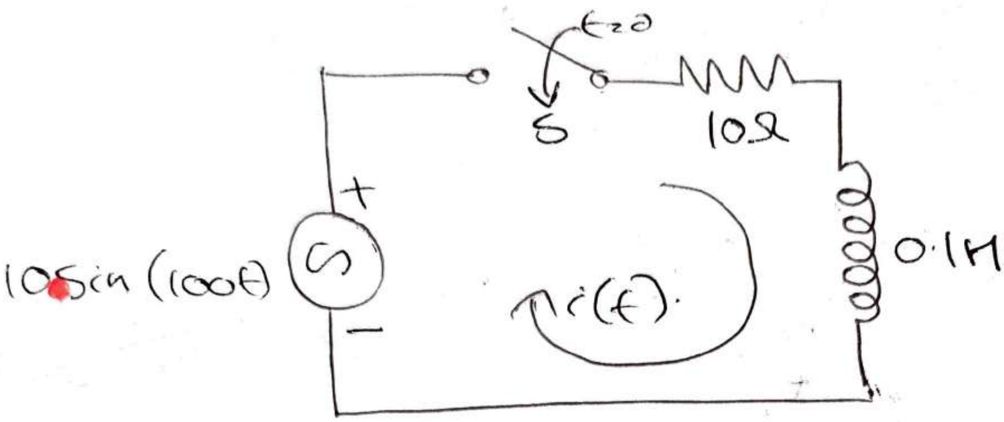


$$= \frac{E_m}{Z^2}\left[\cos\omega t\cdot\sin\phi + \sin\omega t\cdot\cos\phi\right]$$

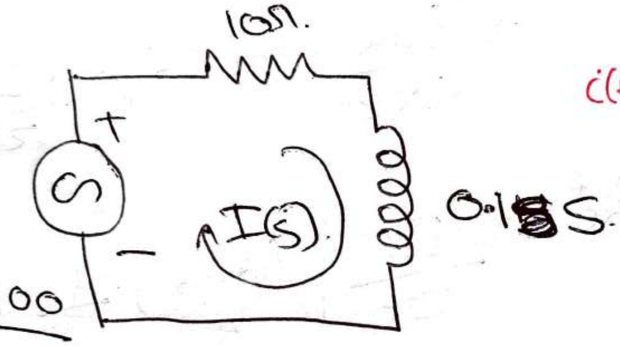
$$i(t) = \underbrace{\frac{-E_m}{\omega C Z^2}\cdot e^{-t/RC}}_{\text{Transcient Part}} + \underbrace{\frac{E_m}{Z^2}\cdot\sin(\omega t+\phi)}_{\text{Steady State Part}}$$

H.W Transcient Analysis of Series RLC circuit excited by AC Source

Q) In the RL circuit shown in Fig, the switch is closed at t=0, find the current i(t) & Voltage across resistance & inductor.



$\Rightarrow$



$i(t) = \frac{V_m}{Z} \sin\left(\omega t + \phi - \tan^{-1}\left(\frac{\omega L}{R}\right)\right)$

$$E(S) = \frac{100 \cdot 100}{S^2+100^2}$$

$$I(S) = \frac{E(S)}{0.1S+10} \Rightarrow \frac{\frac{10^4}{S^2+(10)^4}}{0.1[S+100]} \Rightarrow \frac{10^4}{[S+100][S^2+10^4]}$$

Applying Partial Fraction Technique,

$$\Rightarrow \frac{A}{S+100} + \frac{BS+C}{S^2+10^4}$$

$$A = \left.\frac{10^4}{S^2+10^4}\right|_{S=-100} = \frac{10000}{10^4+10^4} = \frac{1}{2} = \underline{\underline{0.5}}$$

$$B = \underline{\underline{-0.5}}$$

$$10^4 = A[S^2+10^4] + [BS+C][S+100]$$

$100B + C = 0 \Rightarrow C = -100 \times -0.5 = \underline{50}$

$$I(s) = \frac{0.5}{s+100} - \frac{0.5s + 50}{s^2+10^4}$$

$$I(s) = \frac{0.5}{s+100} - \frac{0.5s}{s^2+10^4} + \frac{1}{2} \times \frac{100}{s^2+100^2}$$

$$i(t) = 0.5e^{-100t} - 0.5\cos(100t) + 0.5\sin(100t)$$

$0.5 \Rightarrow 0.5e^{-100t} + \sin(100t)\cdot\cos\phi$

$\cos\phi = \frac{0.5}{0.707} \Rightarrow 0.5 = 0.707\cos\phi$

$\sin\phi = \frac{0.5}{0.707} \Rightarrow 0.5 = 0.707\sin\phi$

$i(t) = 0.5e^{-100t} + \cancel{0.707\cos\phi\cdot\cos(100t) + 0.707\sin\phi\,\sin(100t)}$ $0.707\sin(100t)\cdot\cos\phi - 0.707\cdot\cos(100t)\cdot\sin\phi$

$\boxed{\tan^{-1}\left(\frac{\omega L}{R}\right) = \phi}$

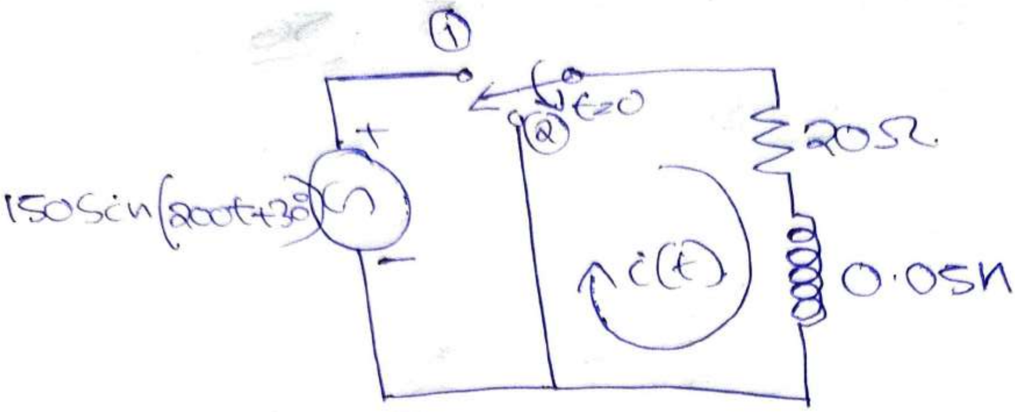
$i(t) = 0.5e^{-100t} + 0.707\cdot\sin(100t - \phi)$

$i(t) = 0.5e^{-100t} + 0.707\sin(100t - 45)$ $\quad \tan\phi = 1$

$\frac{V_m}{Z}\cdot\sin(\omega t + \phi - \tan^{-1}\left(\frac{\omega L}{R}\right))$ $\quad \phi = 45°$

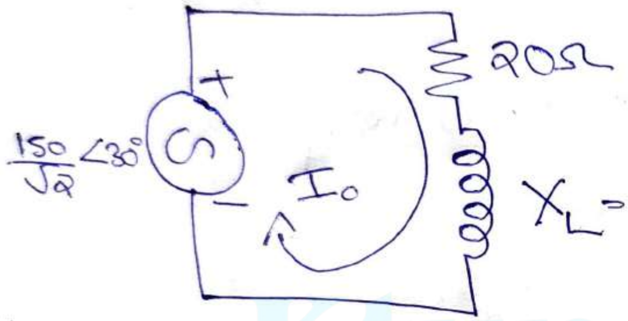
10-10-2020 Saturday

Q) In the circuit, shown in figure the switch remained in position 1 until steady state is reached. At time t=0 the switch is changed to position 2. Find $i(t)$.



Solution) At Position 1

$$\boxed{Z = R + jX_L = R + j\omega L}$$



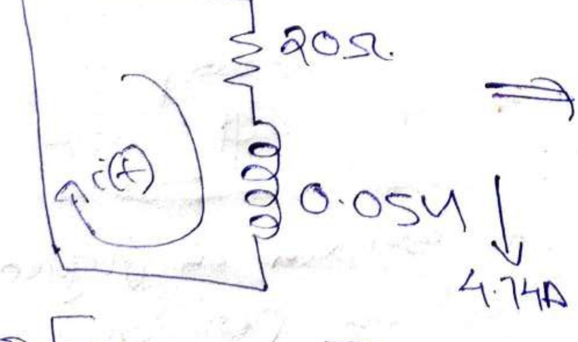
$X_L = j\omega L = 10j$

Let I_o be the magnitude of rms value of current flowing in the circuit.

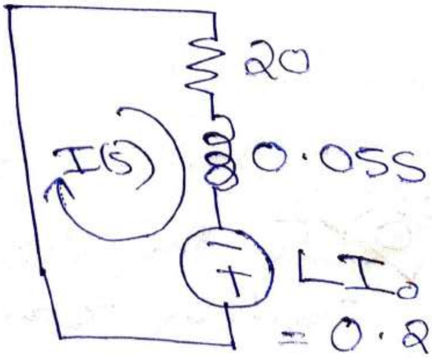
$$|I_o| = \frac{150/\sqrt{2}}{\sqrt{(20)^2 + (10)^2}} \Rightarrow |I_o| = \frac{\frac{150}{\sqrt{2}}}{\sqrt{500}}$$

$|I_o| = \underline{4.74A}$ ⇒ This steady rms current I_o will be the initial current when the switch is moved from position-1 to position-2.

At Position 2



⇒



$= 0.237V$

$$\Rightarrow [20 + 0.05S]\,I(S) = 0.237$$

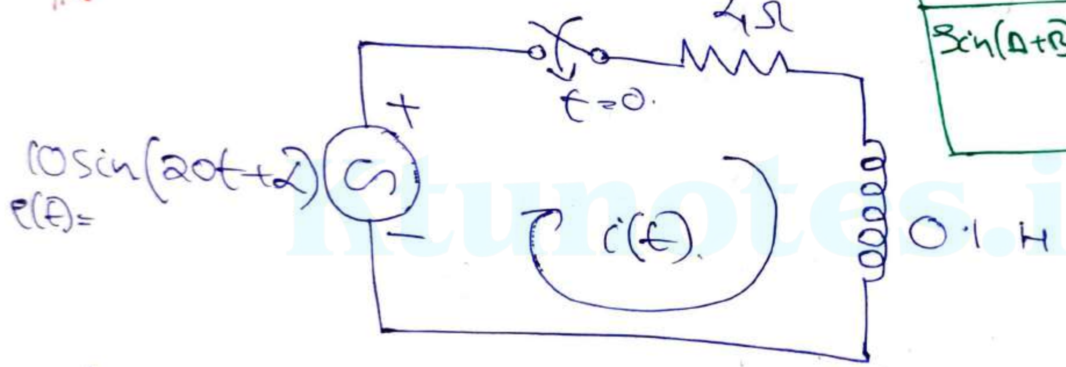
$$0.05\left[S + \frac{20}{0.05}\right] I(S) = 0.237$$

$$I(S) = \frac{4.74}{S+400}$$ ⇒ Now, Taking Inverse Laplace's of.

$$i(t) = 4.74e^{-400t}$$

Q) In the circuit shown in Figure, the switch is closed at a particular value of 'α' so that there is no transient in the RL circuit, find the value of 'α'.

$$\boxed{\sin(A+B) = \sin A \cdot \cos B + \cos A \cdot \sin B}$$

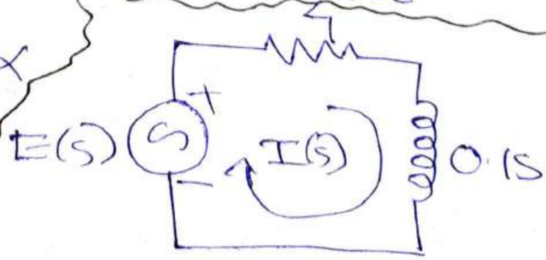


Solution) $E(S) = L\left[10\sin(20t+\alpha)\right]$

$$\Rightarrow 10L\left[\sin 20t \cdot \cos\alpha + \cos 20t \cdot \sin\alpha\right]$$

$$\Rightarrow 10\cos\alpha \times \frac{20}{S^2+400} + 10\sin\alpha \times \frac{S}{S^2+400}$$

$$\Rightarrow \frac{200\cos\alpha + 10S\sin\alpha}{S^2+400}$$



$\Rightarrow \left[4+0.1S\right] I(S) = E(S)$.

$\Rightarrow 0.1\left[S+\frac{4}{0.1}\right] I(S) = E(S)$

$\Rightarrow I(S) = \frac{10E(S)}{S+40}$

$I(S) = \frac{2000\cos\alpha + 100\sin\alpha\, S}{(S+40)(S^2+400)}$

$I(S) = \frac{A}{S+40} + \frac{BS+C}{S^2+400}$.

$A = \left.\frac{2000\cos\alpha + 100\sin\alpha}{S^2+400}\right|_{S=-40}$.

$\cancel{A} = \cancel{2000\cos\alpha + 100\sin\alpha}$ $\quad A = \underline{\cos\alpha - 2\sin\alpha}$

$200\cos\alpha + 100\cdot\sin\alpha \times S$

$\Rightarrow A(S^2+400) + (BS+C)(S+40)$.

Equating the co-efficients of S^2.

$B = -A = -\cos\alpha + 2\sin\alpha$

Equating co-efficients of S

$100\sin\alpha = 40B + C$

$C = 40\cos\alpha + 20\sin\alpha$

$$I(s) = \frac{\cos\alpha - 2\sin\alpha}{s+40} + \frac{(-(\cos\alpha + 2\sin\alpha)s + 40(\cos\alpha + 20\sin\alpha)}{s^2+400}$$

$$I(s) = \frac{A}{s+40} + \frac{B\,s}{s^2+400} + \frac{C \times 20}{20(s^2+400)}$$

$$i(t) = \underbrace{Ae^{-40t}}_{\text{Transcient}} + \underbrace{B\cdot\cos(20t) + \frac{C}{20}\cdot\sin(20t)}_{\text{Steady state}}$$

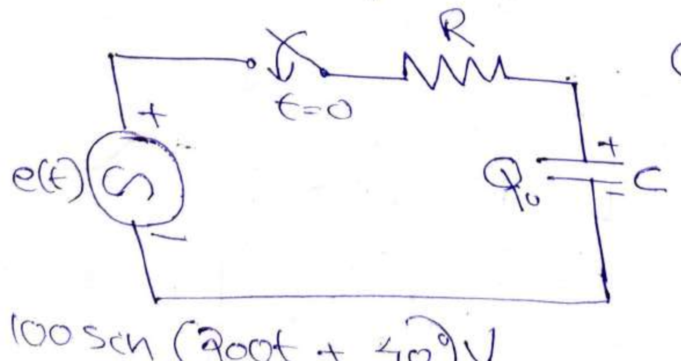
For No transcients in current $i(t)$, the transcient part must be zero.

$Ae^{-40t} = 0$ / At $t=0$, $e^{-40\times 0} = 1$, $\underline{\underline{A=0}}$

ie, $\cos\alpha - 2\sin\alpha = 0 \Rightarrow 2\sin\alpha = \cos\alpha$

$\tan\alpha = 1/2 \Rightarrow \boxed{\alpha = 26.5^\circ}$

Q) For the RC circuit shown in Figure, the switch is closed @ $t=0$. Determine the Voltage across the capacitor.



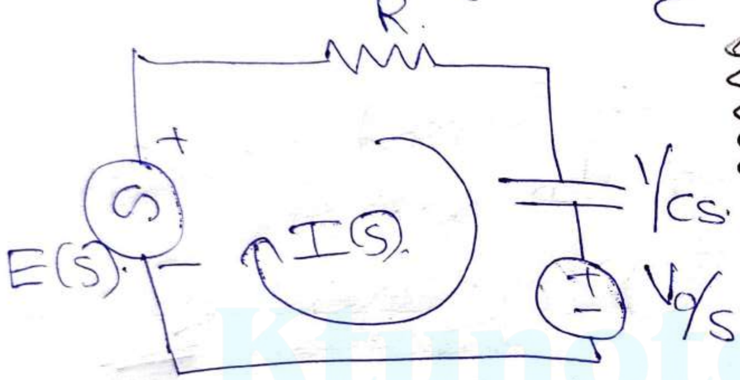
$Q_0 = 250\,\mu C$

$e(t) = 100\sin(200t + 40)\,V$

Solution) $E(S) = 100\cos 40 \times \frac{200}{S^2+200^2} + 100\sin 40 \times \frac{S}{S^2+200^2}$

$$E(S) = \frac{64.2785 + 15320.88}{S^2+(200)^2}$$

Initial Voltage $= \frac{Q_0}{C} = \frac{250\times\text{[illegible]}}{50} = \underline{\underline{5V}}$



$$\left[R + \frac{1}{CS}\right] I(S) = E(S) - \frac{V_0}{S}$$

$$\boxed{I(S) = \frac{E(S) - \frac{V_0}{S}}{R + \frac{1}{CS}}}$$

Voltage across the capacitor

$$V_c(S) = \frac{1}{CS} I(S) + \frac{V_0}{S}$$

$$V_c(S) = \frac{1}{CS}\left[\frac{E(S) - V_0/S}{R + 1/CS}\right] + \frac{V_0}{S}$$

$$V_c(S) = \frac{\cancel{CS}}{\cancel{CS}}\left[\frac{E(S) - V_0/S}{RCS + 1}\right] + \frac{V_0}{S}$$

$$V_c(S) = \frac{1}{RC}\left[\frac{E(S) - V_0/S}{S + 1/RC}\right] + \frac{V_0}{S}$$

$$V_C(S) = \frac{1}{RC}\cdot\frac{E(S)}{S+\frac{1}{RC}} - \frac{V_0/S}{RC\left[S+\frac{1}{RC}\right]} + \frac{V_0}{S}$$

$$V_C(S) = \frac{1}{RC}\cdot\frac{E(S)}{S+\frac{1}{RC}} + \frac{V_0}{S}\left[1 - \frac{1}{RC\left[S+1/RC\right]}\right]$$

$$V_C(S) = \frac{1}{RC}\cdot\frac{E(S)}{S+1/RC} + \frac{V_0}{S}\left[\frac{RC\left[S+\frac{1}{RC}\right]-1}{RC\left[1+1/RC\right]}\right]$$

$$V_C(S) = \frac{V_0}{S}\left[\frac{SRC+1-1}{RC+1}\right] \Rightarrow V_C(S) = \frac{V_0}{S}\times\frac{SRC}{RC\left[1+1/RC\right]}$$

$$V_C(S) = \frac{1}{RC}\cdot\frac{E(S)}{S+1/RC} + \frac{V_0}{S+1/RC}$$

$$V_C(S) = \frac{200(64.2785S+15320.88)}{(S+200)(S^2+(200)^2)} + \frac{5}{S+200}$$

$$V_C(S) = \underbrace{\frac{12855.76S+3064176}{(S+200)(S^2+(200)^2)}} + \frac{5}{S+200}$$

$$V_C(S) = \frac{A}{S+200} + \frac{BS+C}{S^2+(200)^2}$$

$A = 6.1628$, $B = -6.1628$

$C = \underline{14088.37}$

$$V_c(s) = \frac{6.1628}{s+200} + \frac{-6.1628s + 14088.32}{s^2+(200)^2} + \frac{5}{s+200}$$

$$\Rightarrow \frac{6.1628}{s+200} + \frac{5}{s+200} - 6.1628 \times \frac{s}{s^2+(200)^2} +$$

$$\frac{14088.32}{200} \times \frac{200}{s^2+(200)^2}$$

Taking L^{-1}

$$V_c(t) = 11.1628\, e^{-200t} + 6.1628 \cdot \cos(200t) +$$

$$70.4416 \sin(200t)$$

* What is damping ratio?

Solution) The ratio of resistance of a circuit to resistance for critical damping is called damping ratio.